Graphpad Prism 5.0
解析（1）

郑州大学第一附属医院
生物细胞治疗中心
王盟

共享资源

GraphPad 官方网站：

http://www.graphpad.com可下载相关软件，免费试用。

应用简介视频（英语）：

http://www.graphpad.com/prism/prism5movieweb.html

软件简介

What is GraphPad Prism?

是一款集数据分析和作图于一体的数据处理软件，它可以直接输入原始数据，自动进行基本的生物统计，如计算标准差、标准误和 *P*值等，同时产生高质量的科学图表。

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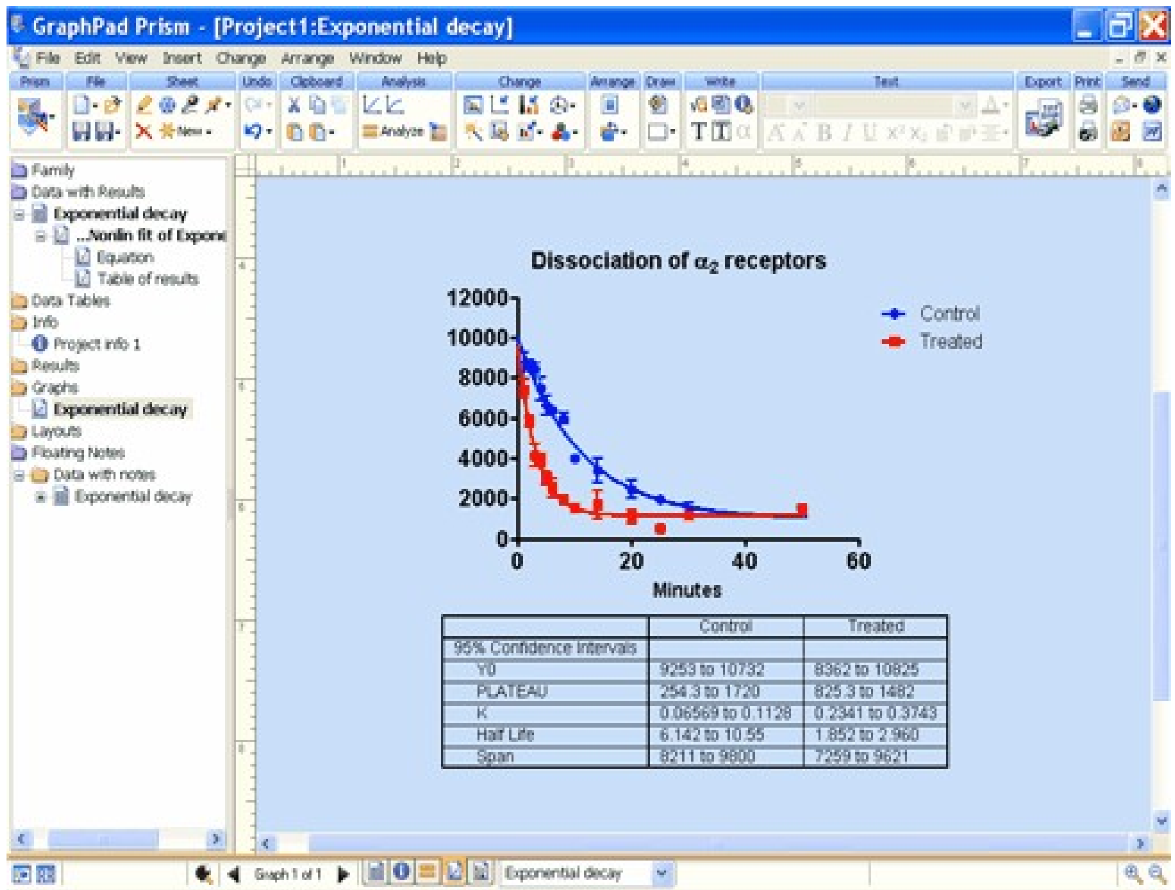
图
是这款软件的
核心。



软件自带的学习资料

- **Movies**

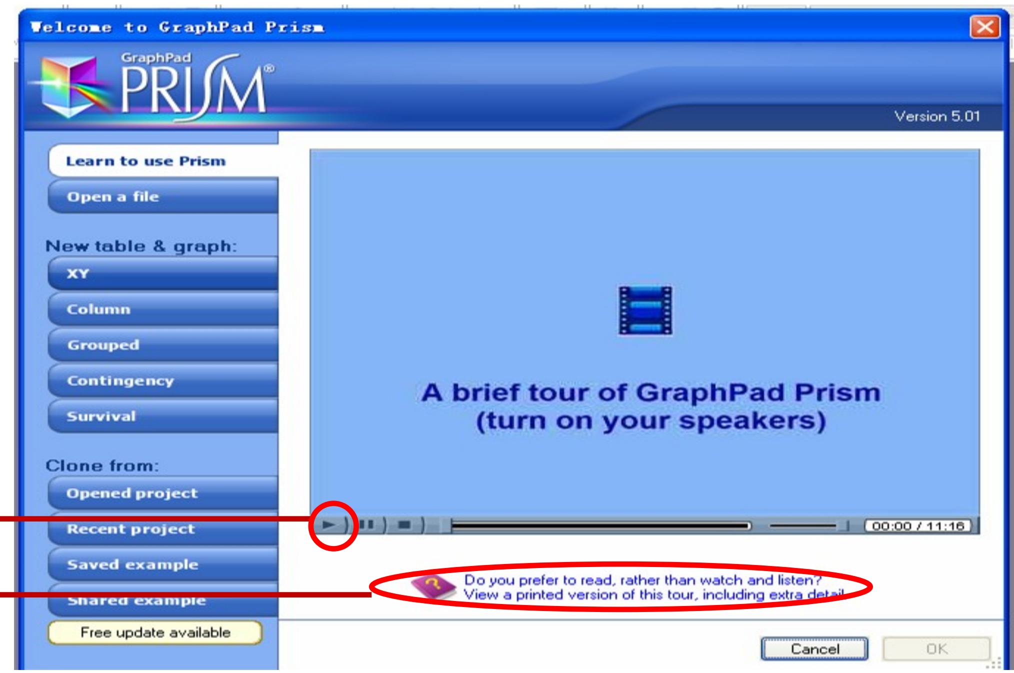
最舒服的一种方式。

- **详细使用说明**

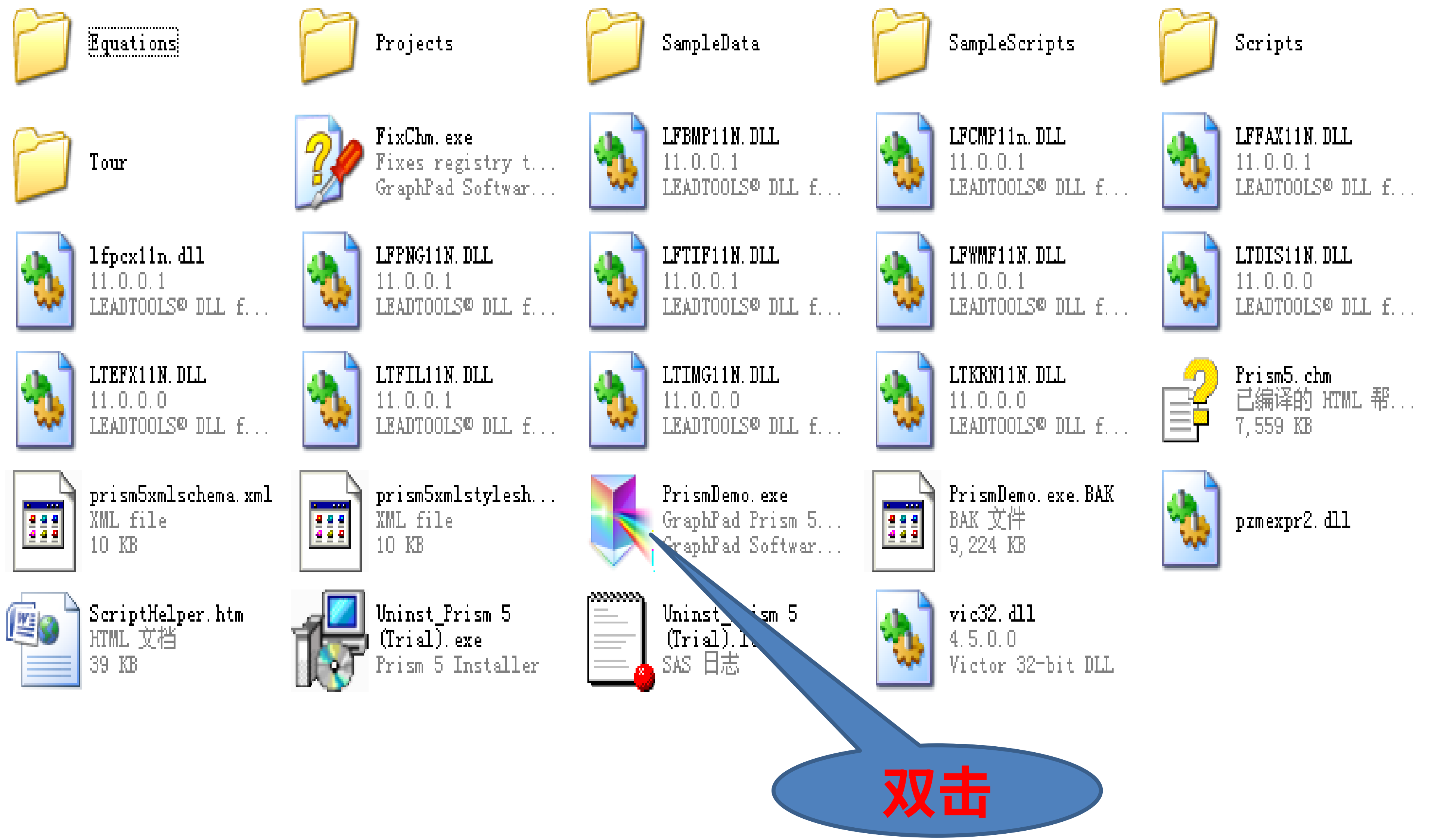
最详细，内容多需花大力气研究学习。

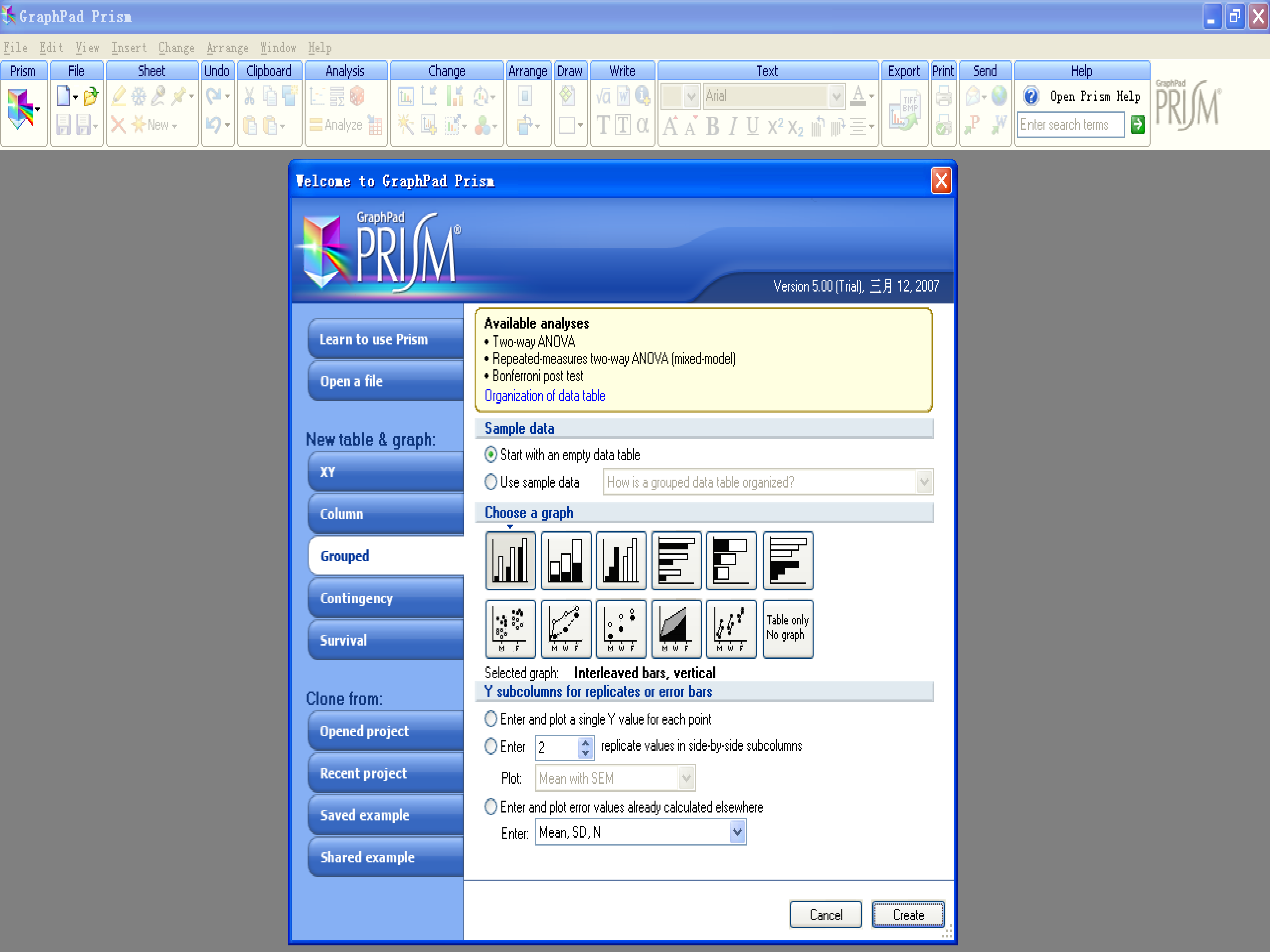
- **快速入门手册**

快速掌握基本技能知识和新特点。

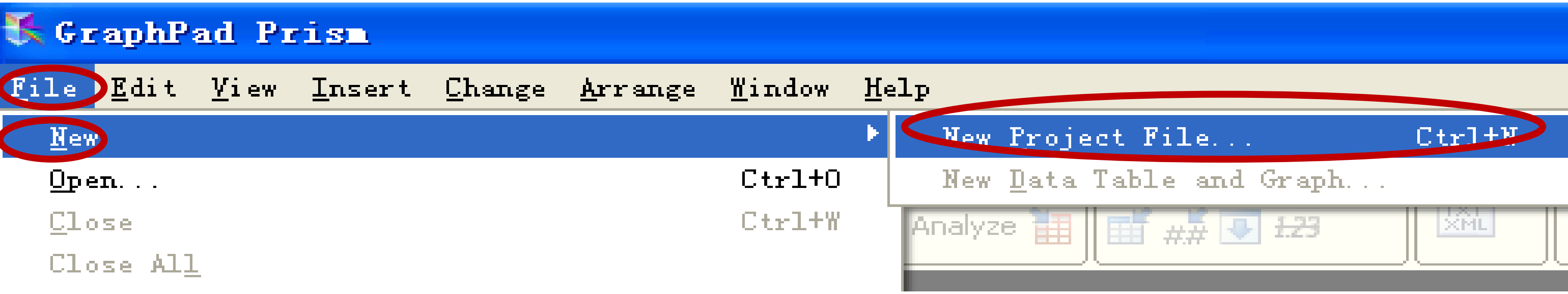


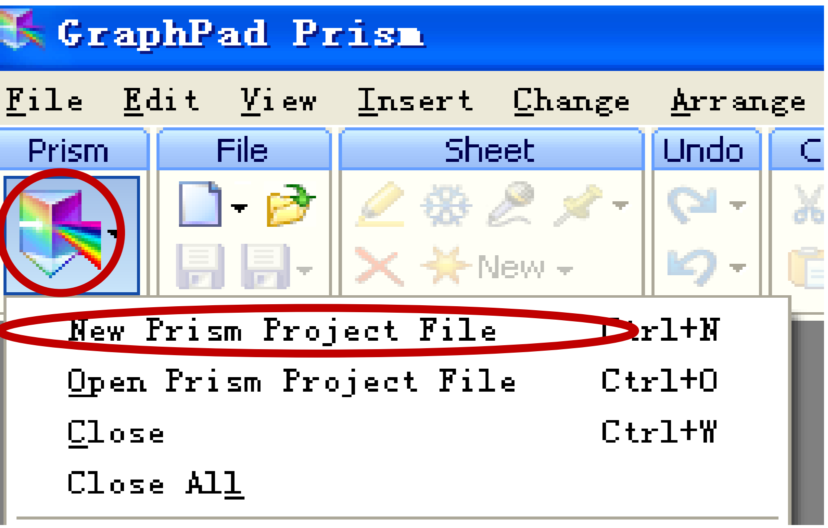
一切从头开始

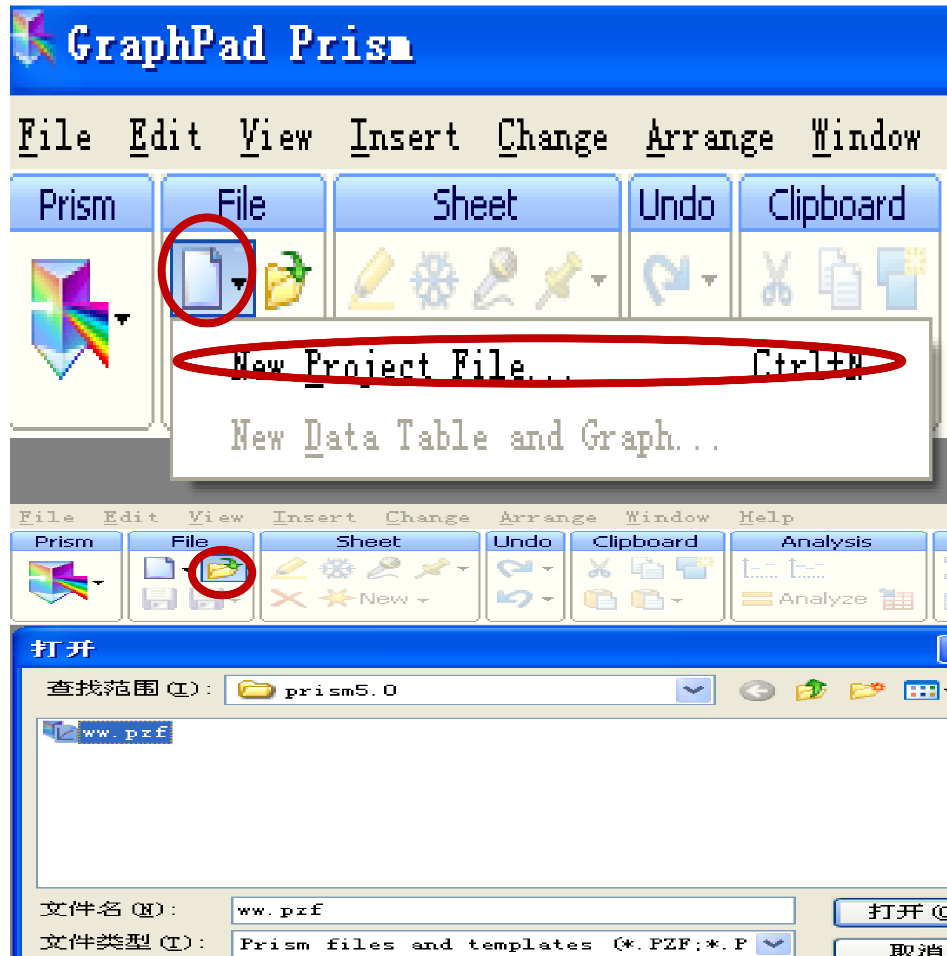

Equations
Projects
SampleData
SampleScripts
Scripts
Tour
FixChm.exe
Fixes registry t...
GraphPad Softwar...
LFBMP11N.DLL
11.0.0.1
LEADTOOLS® DLL f...
LFCMP11n.DLL
11.0.0.1
LEADTOOLS® DLL f...
LFFAX11N.DLL
11.0.0.1
LEADTOOLS® DLL f...
lfpcx11n.dll
11.0.0.1
LEADTOOLS® DLL f...
LFPNG11N.DLL
11.0.0.1
LEADTOOLS® DLL f...
LFTIF11N.DLL
11.0.0.1
LEADTOOLS® DLL f...
LFWMF11N.DLL
11.0.0.1
LEADTOOLS® DLL f...
LTDIS11N.DLL
11.0.0.0
LEADTOOLS® DLL f...
LTEFX11N.DLL
11.0.0.0
LEADTOOLS® DLL f...
LTFIL11N.DLL
11.0.0.1
LEADTOOLS® DLL f...
LTIMG11N.DLL
11.0.0.0
LEADTOOLS® DLL f...
LTKRN11N.DLL
11.0.0.0
LEADTOOLS® DLL f...
Prism5.chm
已编译的 HTML 帮...
7,559 KB
prism5xmlschema.xml
XML file
10 KB
prism5xmlstylesh...
XML file
10 KB
PrismDemo.exe
GraphPad Prism 5...
GraphPad Softwar...
PrismDemo.exe.BAK
BAK 文件
9,224 KB
pzmexpr2.dll
ScriptHelper.htm
HTML 文档
39 KB
Uninst_Prism 5
(Trial).exe
Prism 5 Installer
SAS 日志
vic32.dll
4.5.0.0
Victor 32-bit DLL
双击

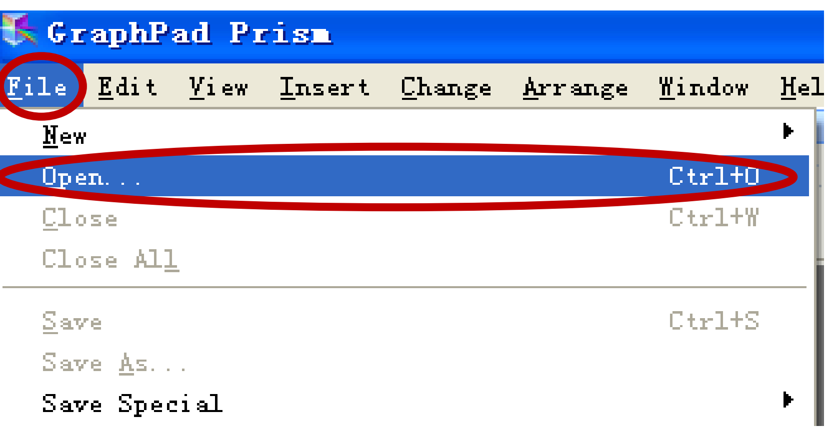

GraphPad Prism
File Edit View Insert Change Arrange Window Help
Prism File Sheet Undo Clipboard Analysis Change Arrange Draw Write Text Export Print Send Help
Analyze
New
Arial
Open Prism Help
Enter search terms
Welcome to GraphPad Prism
GraphPad PRISM
Version 5.00 (Trial), 三月 12, 2007
Learn to use Prism
Open a file
New table & graph:
XY
Column
Grouped
Contingency
Survival
Clone from:
Opened project
Recent project
Saved example
Shared example
Available analyses
• Two-way ANOVA
• Repeated-measures two-way ANOVA (mixed-model)
• Bonferroni post test
Organization of data table
Sample data
Start with an empty data table
Use sample data
How is a grouped data table organized?
Choose a graph
Table only No graph
Selected graph: Interleaved bars, vertical
Y subcolumns for replicates or error bars
Enter and plot a single Y value for each point
Enter 2 replicate values in side-by-side subcolumns
Plot: Mean with SEM
Enter and plot error values already calculated elsewhere
Enter: Mean, SD, N
Cancel
Create

新建文件及打开已有文件的方法

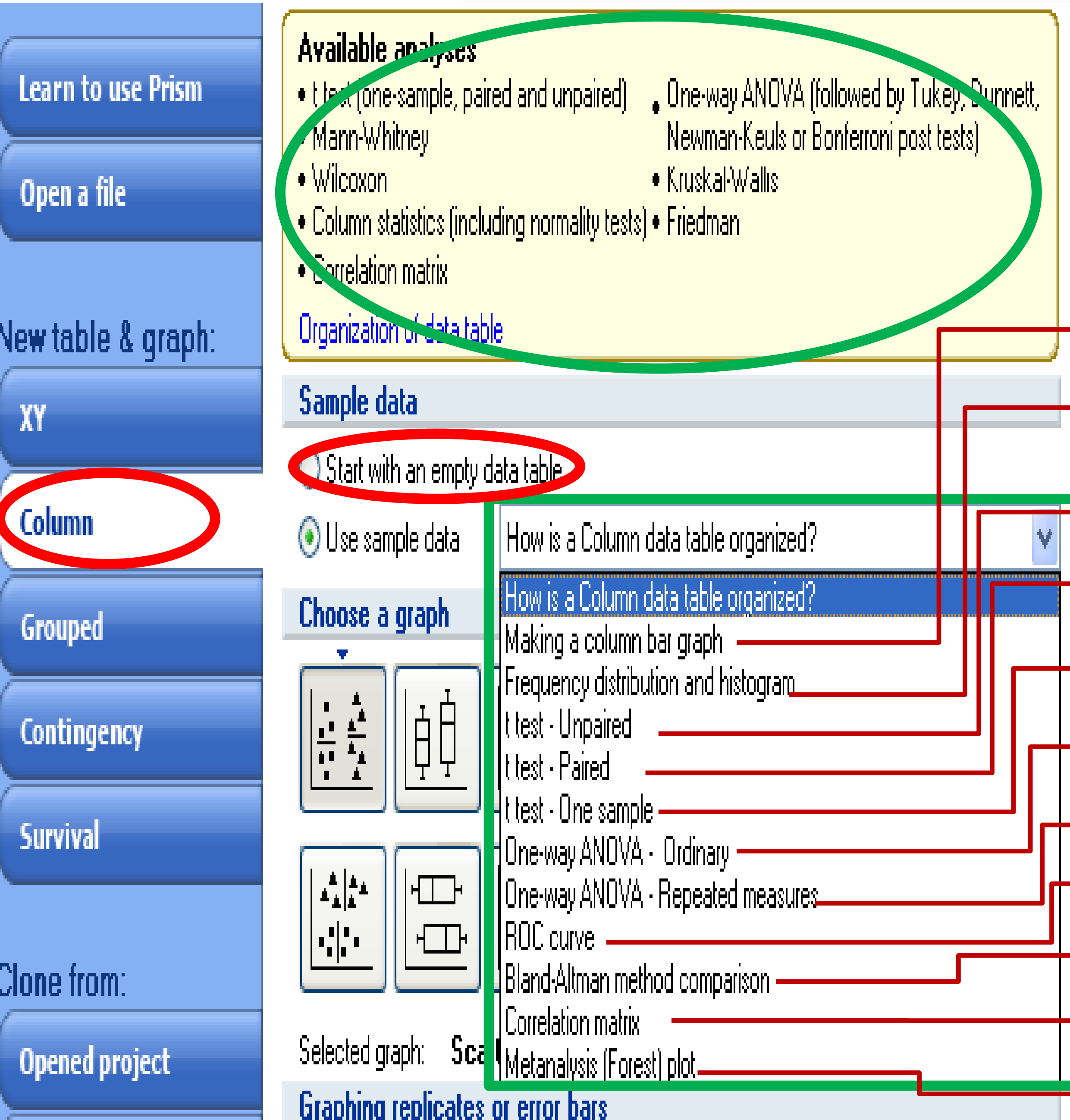








Column 分析模块



以列变量做柱状图

以列变量做频数分布或柱状图

非配对 t 检验

配对 t 检验

单样本 t 检验

单因素方差分析

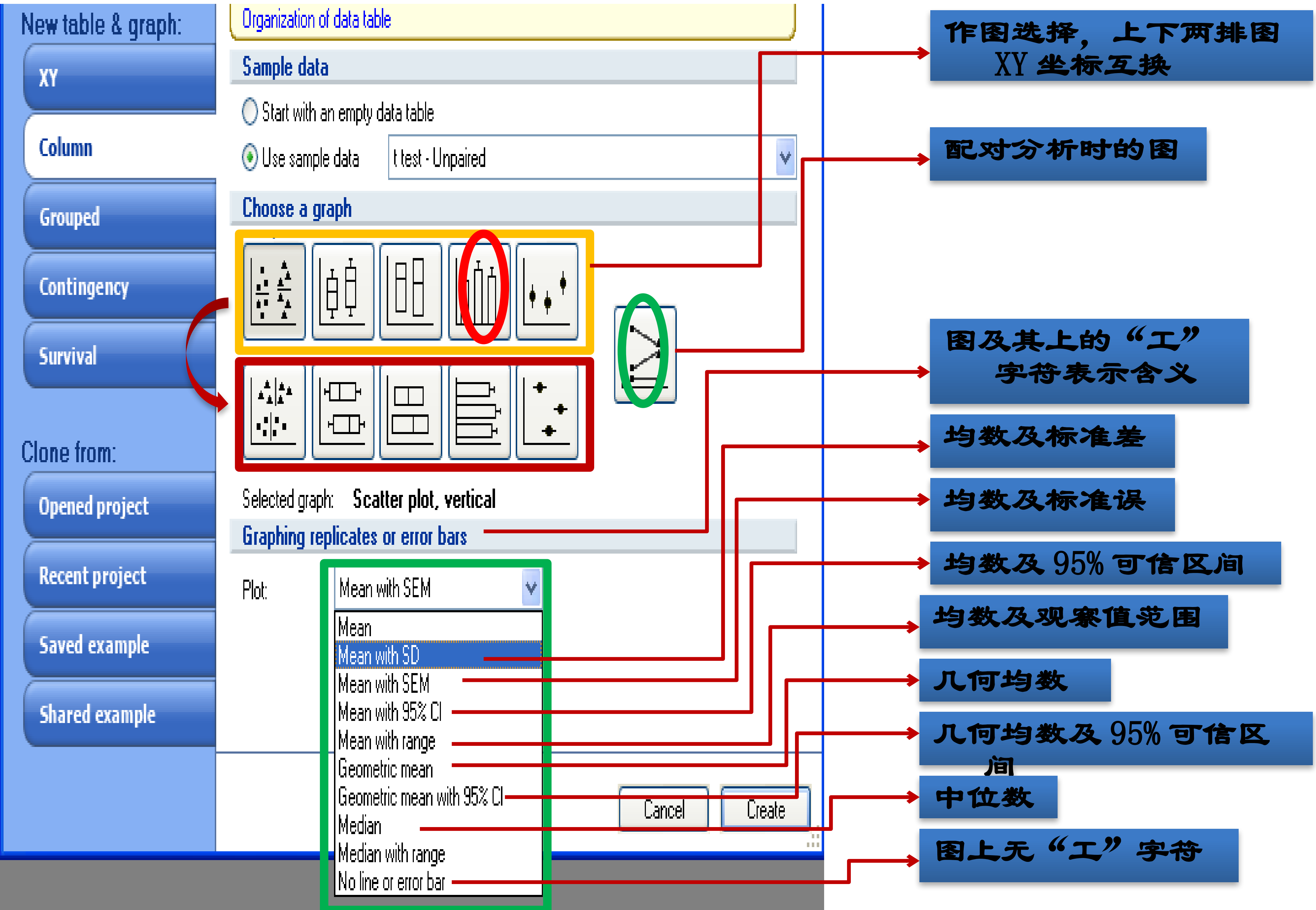
重复测量单因素方差分析

感受性曲线

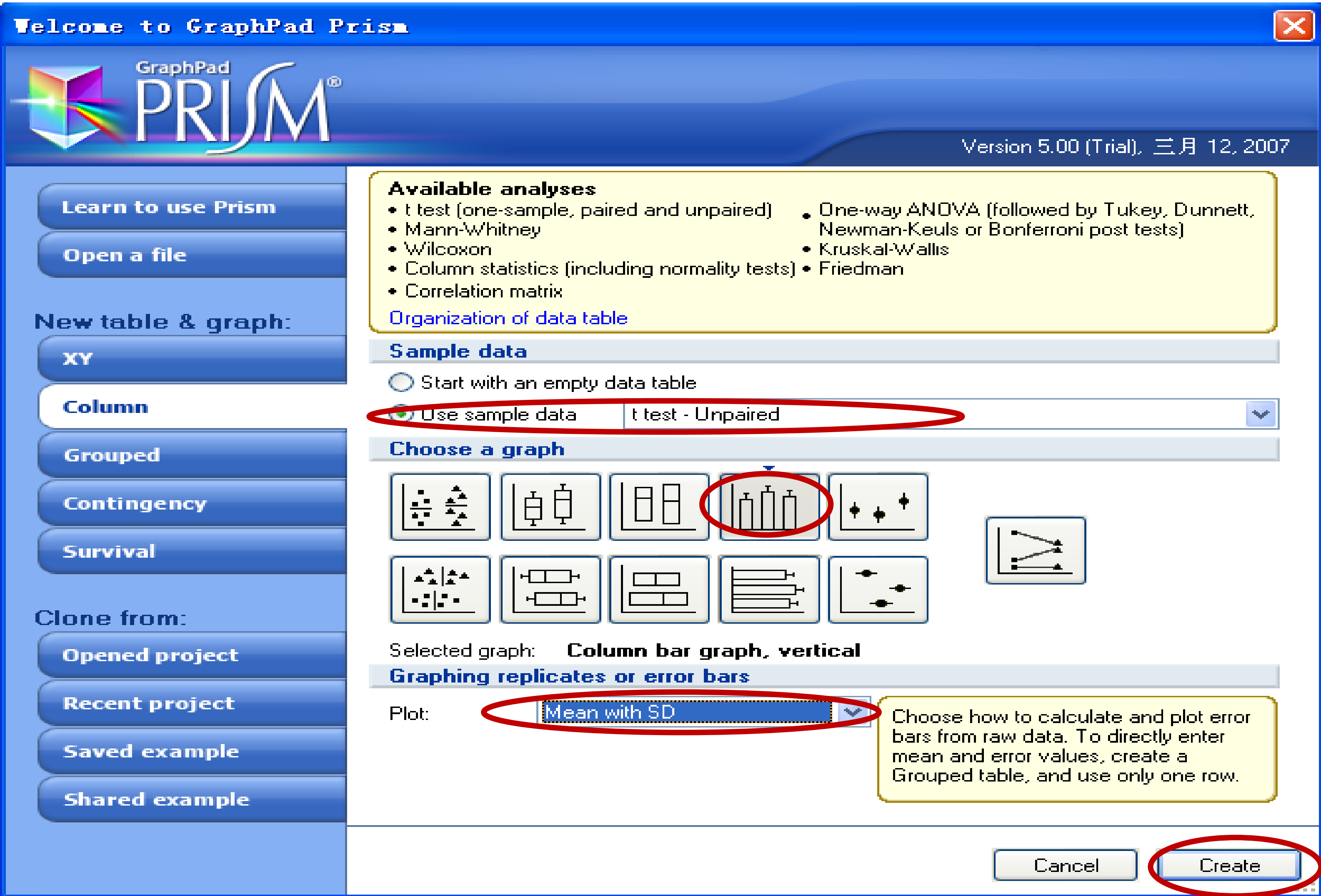
两种方法测量一致性检验

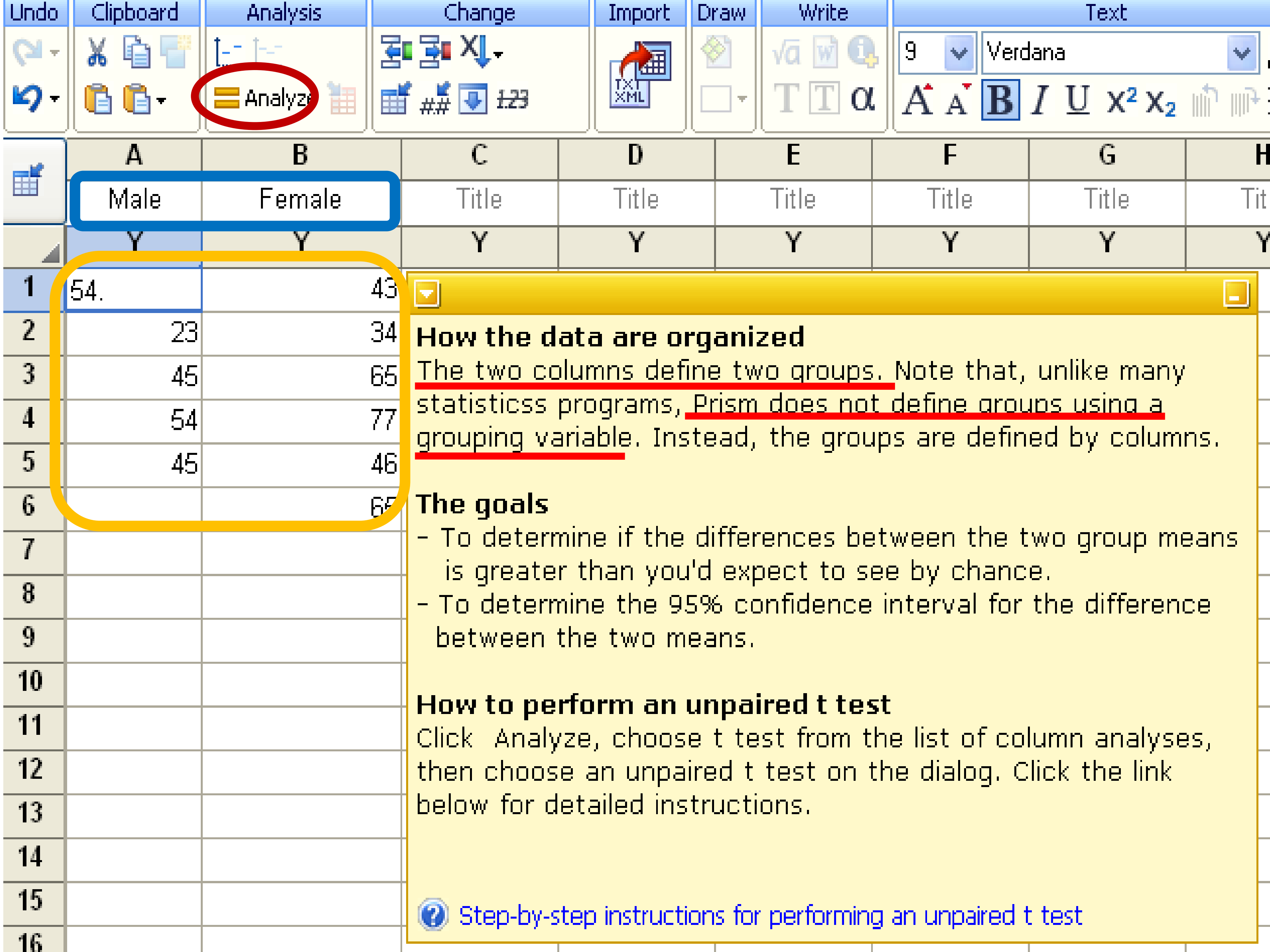
相关矩阵

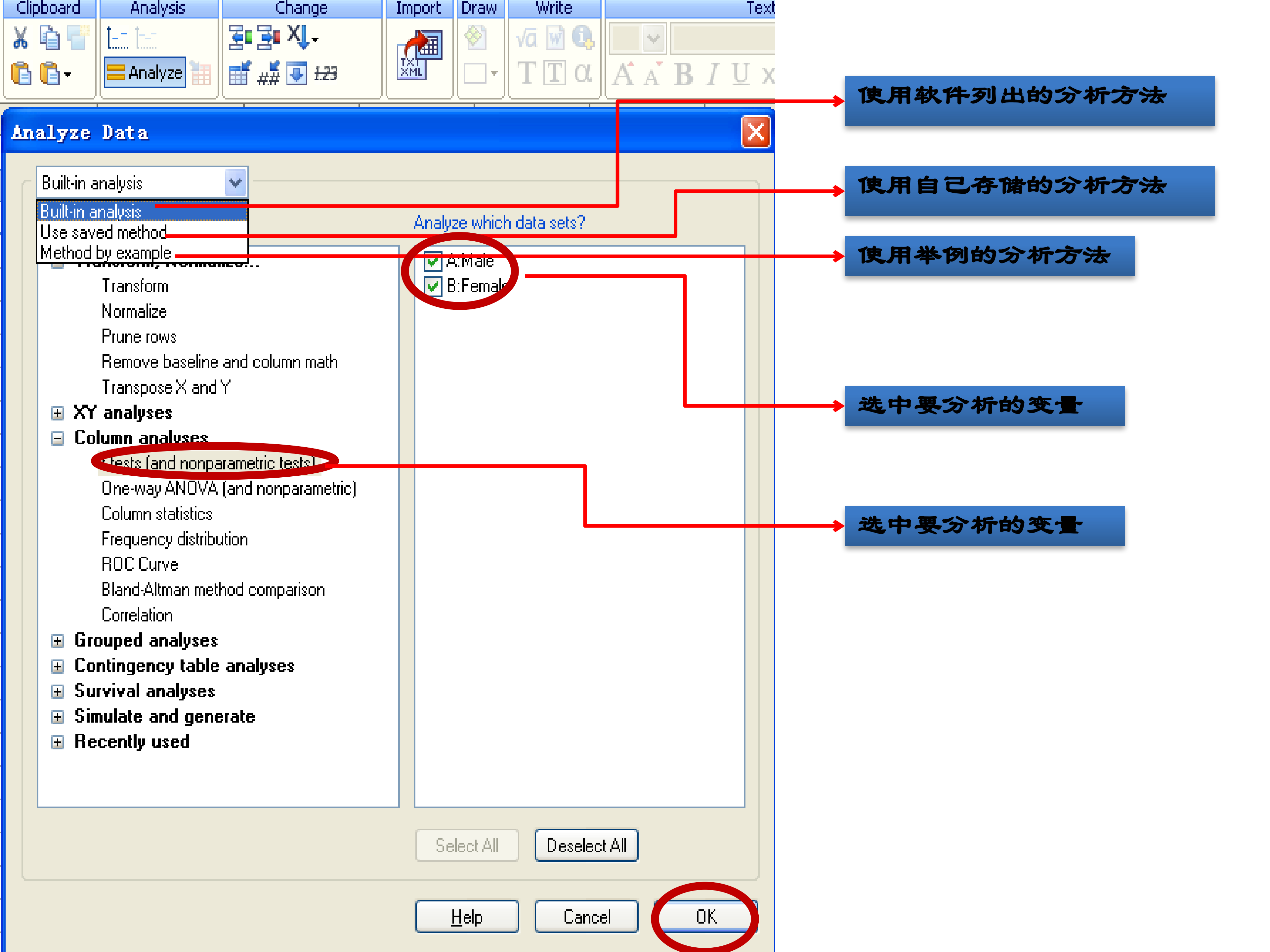
Met 分析


New table & graph:
XY
Column
Grouped
Contingency
Survival
Clone from:
Opened project
Recent project
Saved example
Shared example
Organization of data table
Sample data
Start with an empty data table
Use sample data
t test - Unpaired
Choose a graph
Selected graph: Scatter plot, vertical
Graphing replicates or error bars
Plot:
Mean with SEM
Mean
Mean with SD
Mean with SEM
Mean with 95% CI
Mean with range
Geometric mean
Geometric mean with 95% CI
Median
Median with range
No line or error bar
Cancel
Create
作图选择，上下两排图XY坐标互换
配对分析时的图
图及其上的“工”字符表示含义
均数及标准差
均数及标准误
均数及95%可信区间
均数及观察值范围
几何均数
几何均数及95%可信区间
中位数
图上无“工”字符

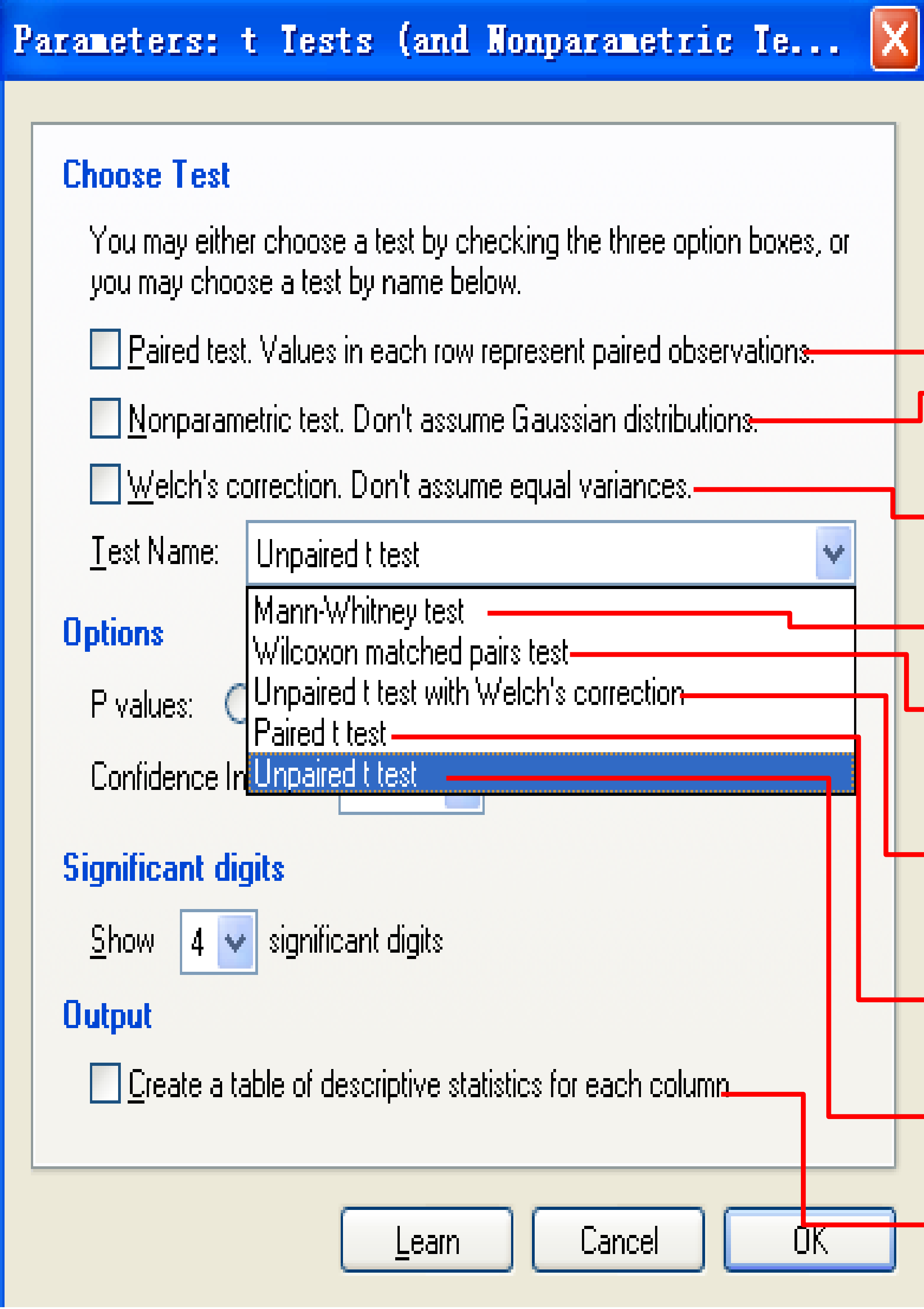
非配对 t 检验


Welcome to GraphPad Prism
GraphPad PRISM
Version 5.00 (Trial), 三月 12, 2007
Learn to use Prism
Open a file
New table & graph:
XY
Column
Grouped
Contingency
Survival
Clone from:
Opened project
Recent project
Saved example
Shared example
Available analyses
• t test (one-sample, paired and unpaired)
• Mann-Whitney
• Wilcoxon
• Column statistics (including normality tests)
• Correlation matrix
• One-way ANOVA (followed by Tukey, Dunnett, Newman-Keuls or Bonferroni post tests)
• Kruskal-Wallis
• Friedman
Organization of data table
Sample data
Start with an empty data table
Use sample data
t test - Unpaired
Choose a graph
Selected graph: Column bar graph, vertical
Graphing replicates or error bars
Plot:
Mean with SD
Choose how to calculate and plot error bars from raw data. To directly enter mean and error values, create a Grouped table, and use only one row.
Cancel
Create


Undo
Clipboard
Analysis
Change
Import
Draw
Write
Text
Analyze
TXT XML
9
Verdana
A
B
C
D
E
F
G
H
Male
Female
Title
Title
Title
Title
Title
Tit
Y
Y
Y
Y
Y
Y
Y
Y
1 54. 43
2 23 34
3 45 65
4 54 77
5 45 46
6 65
7
8
9
10
11
12
13
14
15
16
How the data are organized
The two columns define two groups. Note that, unlike many statisticss programs, Prism does not define groups using a grouping variable. Instead, the groups are defined by columns.
The goals
- To determine if the differences between the two group means is greater than you'd expect to see by chance.
- To determine the 95% confidence interval for the difference between the two means.
How to perform an unpaired t test
Click Analyze, choose t test from the list of column analyses, then choose an unpaired t test on the dialog. Click the link below for detailed instructions.
Step-by-step instructions for performing an unpaired t test



Clipboard
Analysis
Change
Import
Draw
Write
Text
Analyze
Analyze Data
Built-in analysis
Built-in analysis
Use saved method
Method by example
Analyze which data sets?
A:Male
B:Female
Transform
Normalize
Prune rows
Remove baseline and column math
Transpose X and Y
XY analyses
Column analyses
t tests (and nonparametric tests)
One-way ANOVA (and nonparametric)
Column statistics
Frequency distribution
ROC Curve
Bland-Altman method comparison
Correlation
Grouped analyses
Contingency table analyses
Survival analyses
Simulate and generate
Recently used
Select All
Deselect All
Help
Cancel
OK
使用软件列出的分析方法
使用自己存储的分析方法
使用举例的分析方法
选中要分析的变量
选中要分析的变量



资料符合正态分布时的配对检验

资料不符合正态分布时的非参数检验

资料方差不齐时 Welch’s 校准后独立样本均数比较的秩和检验

非参检验，两独立样本检验

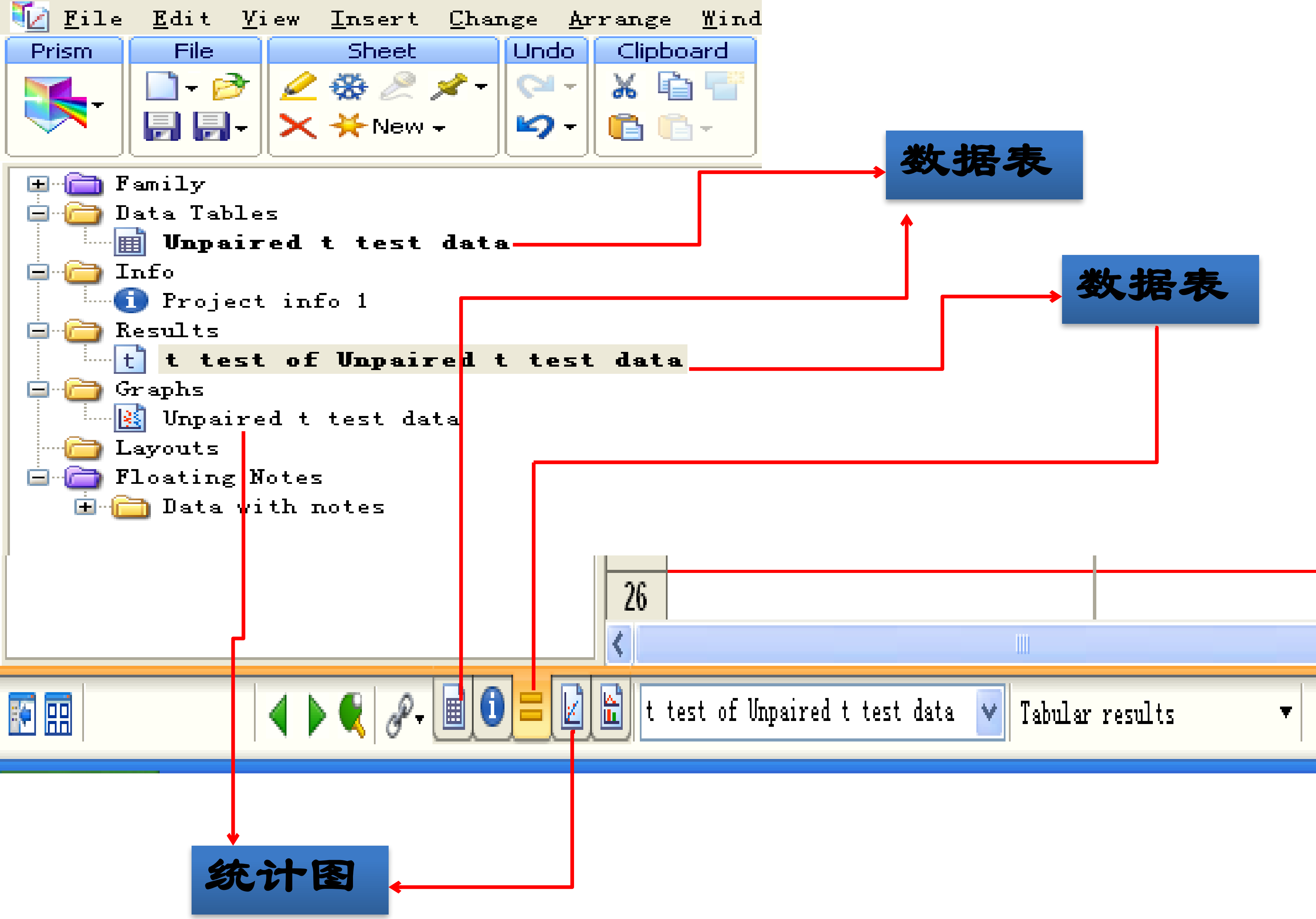
非参检验，两相关样本检验（配对、配伍）

用 Welch’s 校准的两样本非配对 t 检验

正态分布资料配对样本均数比较 t 检验

正态分布资料非配对样本均数比较 t 检验

对每列资料进行描述统计分析


File Edit View Insert Change Arrange Wind
Prism
File
Sheet
Undo
Clipboard
New
Family
Data Tables
Unpaired t test data
Info
Project info 1
Results
t test of Unpaired t test data
Graphs
Unpaired t test data
Layouts
Floating Notes
Data with notes
26
t test of Unpaired t test data
Tabular results
数据表
数据表
统计图

t test
Tabular results

t 检验结果

1	Table Analyzed	Unpaired t test data
2	Column A	Male
3	vs	vs
4	Column B	Female
5		
6	Unpaired t test	
7	P value	0.2613
8	P value summary	ns
9	Are means signif. different? (P < 0.05)	No
10	One- or two-tailed P value?	Two-tailed
11	t, df	t=1.199 df=9
12		
13	How big is the difference?	
14	Mean ± SEM of column A	44.20 ?5.669 N=5
15	Mean ± SEM of column B	55.00 ?6.708 N=6
16	Difference between means	-10.80 ?9.010
17	95% confidence interval	-31.18 to 9.582
18	R squared	0.1377
19		
20	F test to compare variances	
21	F,DFn, Dfd	1.680, 5, 4
22	P value	0.6354
23	P value summary	ns
24	Are variances significantly different?	No

统计方法

P值

统计结果判定

t值

自由度

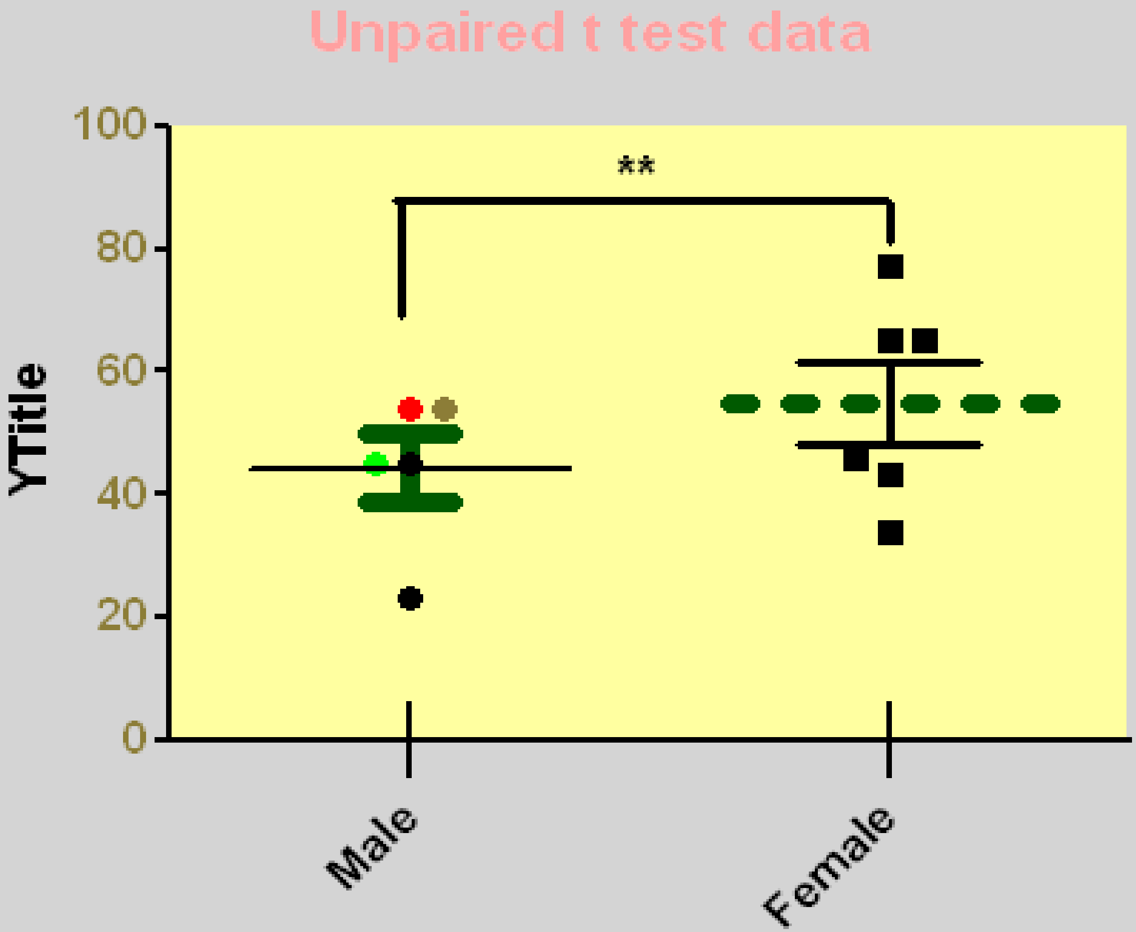
两组方差齐性检验

两组方差齐

t test Column statistics	A Male Y	B Female Y
1 Number of values	5	6
2 Minimum	23.00	34.00
3 25% Percentile	34.00	40.75
4 Median	45.00	55.50
5 75% Percentile	54.00	68.00
6 Maximum	54.00	77.00
7		
8 Mean	44.20	55.00
9 Std. Deviation	12.68	16.43
10 Std. Error	5.669	6.708
11		
12 Lower 95% CI	28.46	37.76
13 Upper 95% CI	59.94	72.24

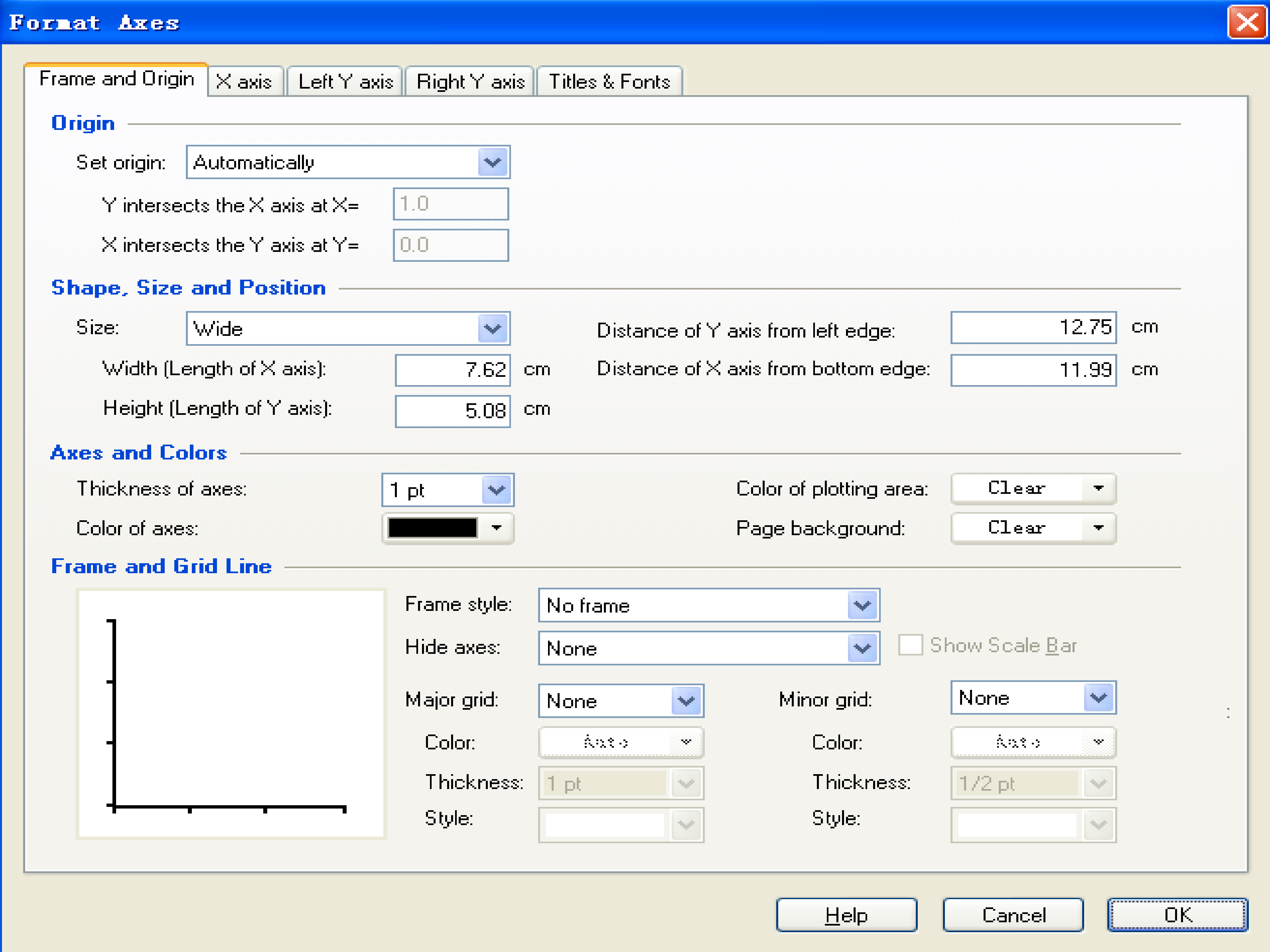
列变量特征

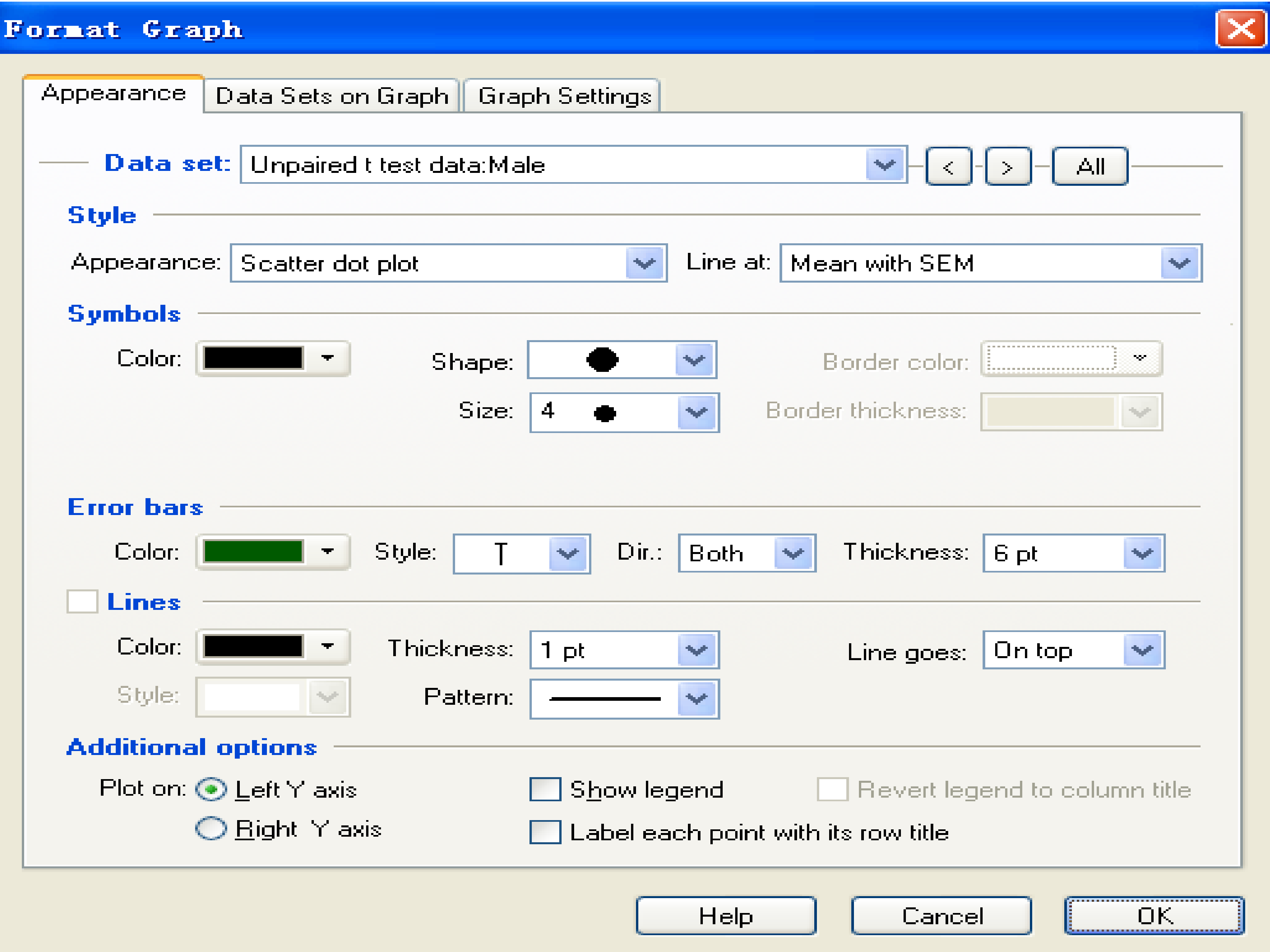
均数、标准差、标准误



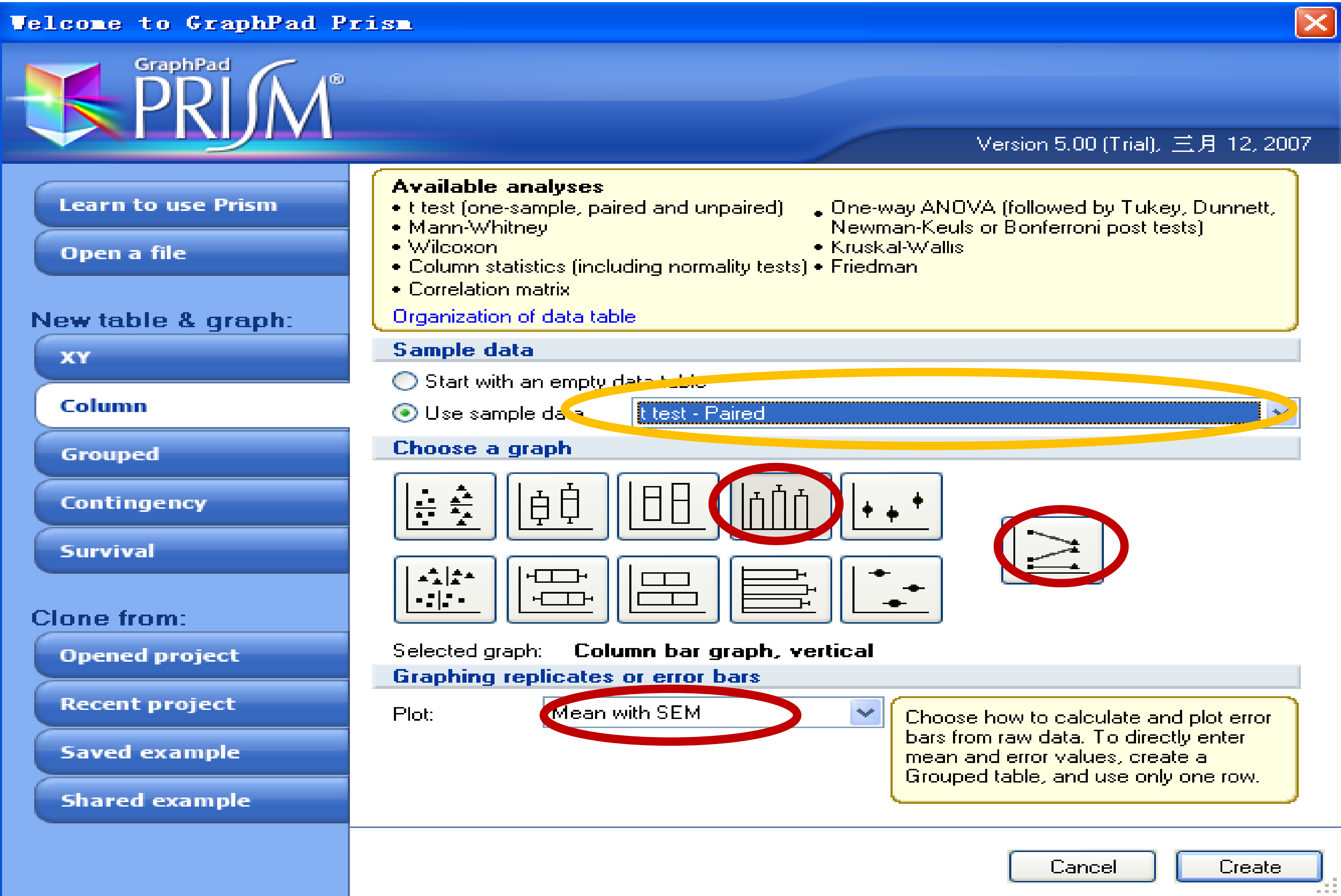
男性	女性
54	43
23	34
45	65
54	77
45	46
	65

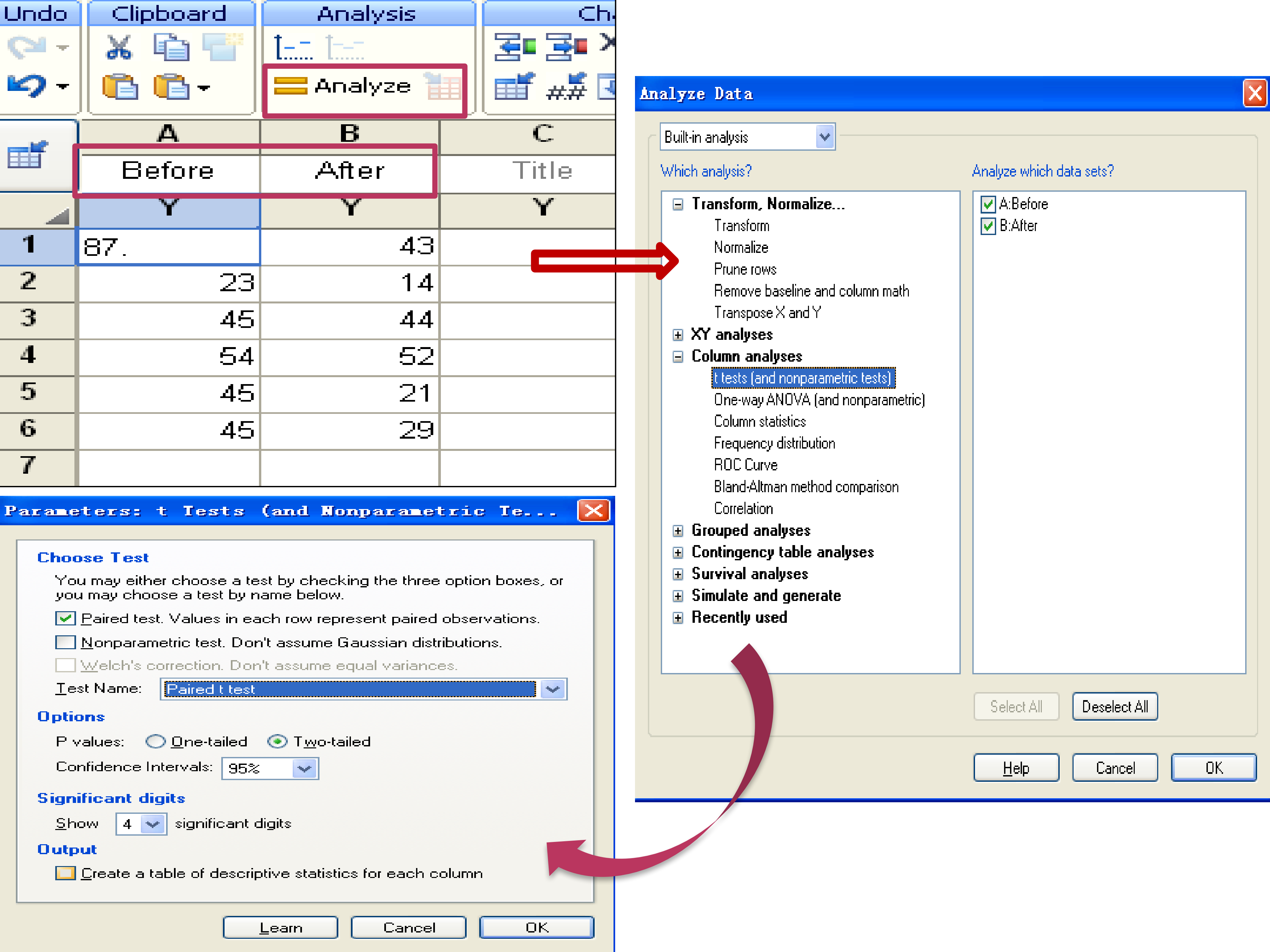
Figure 2. Proportions of T cell receptor (TCR) T cells and NK cells in healthy donors (HDs), asymptomatic carriers (AsCs) of hepatitis B virus, patients with low-grade chronic hepatitis B (CHB-L), and patients with moderate or severe chronic hepatitis B (CHB-M/S). The distribution statistics


Format Axes
Frame and Origin
X axis
Left Y axis
Right Y axis
Titles & Fonts
Origin
Set origin:
Automatically
Y intersects the X axis at X=
1.0
X intersects the Y axis at Y=
0.0
Shape, Size and Position
Size:
Wide
Width (Length of X axis):
7.62
cm
Height (Length of Y axis):
5.08
cm
Distance of Y axis from left edge:
12.75
cm
Distance of X axis from bottom edge:
11.99
cm
Axes and Colors
Thickness of axes:
1 pt
Color of axes:
Color of plotting area:
Clear
Page background:
Clear
Frame and Grid Line
Frame style:
No frame
Hide axes:
None
Show Scale Bar
Major grid:
None
Color:
Thickness:
1 pt
Style:
Minor grid:
None
Color:
Thickness:
1/2 pt
Style:
Help
Cancel
OK


Format Graph
Appearance
Data Sets on Graph
Graph Settings
Data set:
Unpaired t test data:Male
<
>
All
Style
Appearance:
Scatter dot plot
Line at:
Mean with SEM
Symbols
Color:
Shape:
Border color:
Size:
4
Border thickness:
Error bars
Color:
Style:
Dir.:
Both
Thickness:
6 pt
Lines
Color:
Thickness:
1 pt
Line goes:
On top
Style:
Pattern:
Additional options
Plot on:
Left Y axis
Right Y axis
Show legend
Revert legend to column title
Label each point with its row title
Help
Cancel
OK

配对 *t* 检验


Welcome to GraphPad Prism
GraphPad PRISM®
Version 5.00 (Trial), 三月 12, 2007
Learn to use Prism
Open a file
New table & graph:
XY
Column
Grouped
Contingency
Survival
Clone from:
Opened project
Recent project
Saved example
Shared example
Available analyses
• t test (one-sample, paired and unpaired)
• Mann-Whitney
• Wilcoxon
• Column statistics (including normality tests)
• Correlation matrix
• One-way ANOVA (followed by Tukey, Dunnett, Newman-Keuls or Bonferroni post tests)
• Kruskal-Wallis
• Friedman
Organization of data table
Sample data
Start with an empty data table
Use sample data
t test - Paired
Choose a graph
Selected graph: Column bar graph, vertical
Graphing replicates or error bars
Plot:
Mean with SEM
Choose how to calculate and plot error bars from raw data. To directly enter mean and error values, create a Grouped table, and use only one row.
Cancel
Create

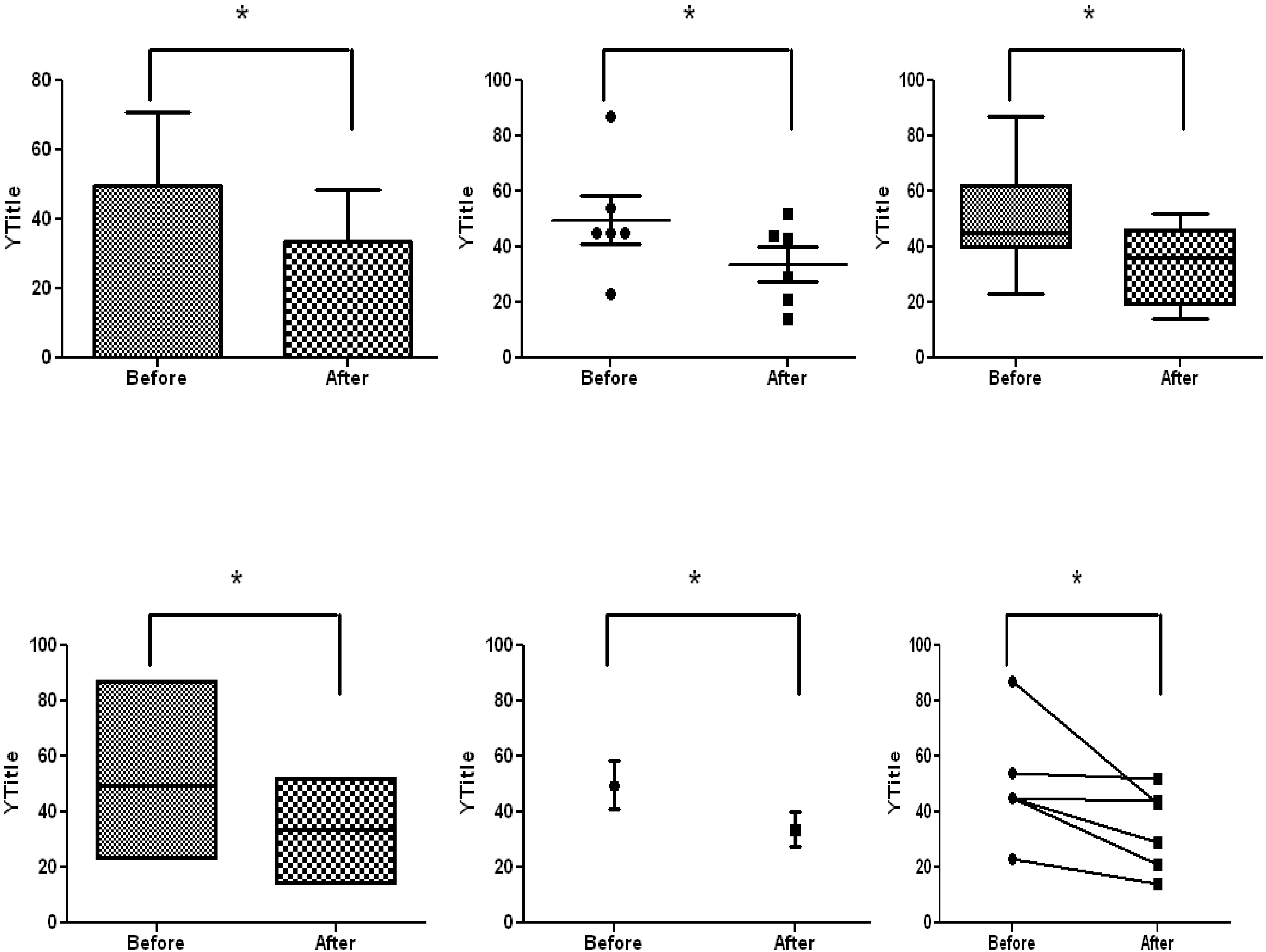

Undo
Clipboard
Analysis
Analyze
A
B
C
Before
After
Title
Y
Y
Y
1
87.
43
2
23
14
3
45
44
4
54
52
5
45
21
6
45
29
7
Analyze Data
Built-in analysis
Which analysis?
Analyze which data sets?
Transform, Normalize...
Transform
Normalize
Prune rows
Remove baseline and column math
Transpose X and Y
XY analyses
Column analyses
t tests (and nonparametric tests)
One-way ANOVA (and nonparametric)
Column statistics
Frequency distribution
ROC Curve
Bland-Altman method comparison
Correlation
Grouped analyses
Contingency table analyses
Survival analyses
Simulate and generate
Recently used
A:Before
B:After
Select All
Deselect All
Help
Cancel
OK
Parameters: t Tests (and Nonparametric Te...
Choose Test
You may either choose a test by checking the three option boxes, or you may choose a test by name below.
Paired test. Values in each row represent paired observations.
Nonparametric test. Don't assume Gaussian distributions.
Welch's correction. Don't assume equal variances.
Test Name:
Paired t test
Options
P values:
One-tailed
Two-tailed
Confidence Intervals:
95%
Significant digits
Show
4
significant digits
Output
Create a table of descriptive statistics for each column
Learn
Cancel
OK

t test

1	Table Analyzed	Paired t test data
2	Column A	Before
3	vs	vs
4	Column B	After
5		
6	Paired t test	
7	P value	0.0606
8	P value summary	ns
9	Are means signif. different? (P < 0.05)	No
10	One- or two-tailed P value?	Two-tailed
11	t, df	t=2.414 df=5
12	Number of pairs	6
13		
14	How big is the difference?	
15	Mean of differences	16.00
16	95% confidence interval	-1.041 to 33.04
17	R squared	0.5382
18		
19	How effective was the pairing?	
20	Correlation coefficient (r)	0.6350
21	P Value (one tailed)	0.0878
22	P value summary	ns
23	Was the pairing significantly effective?	No

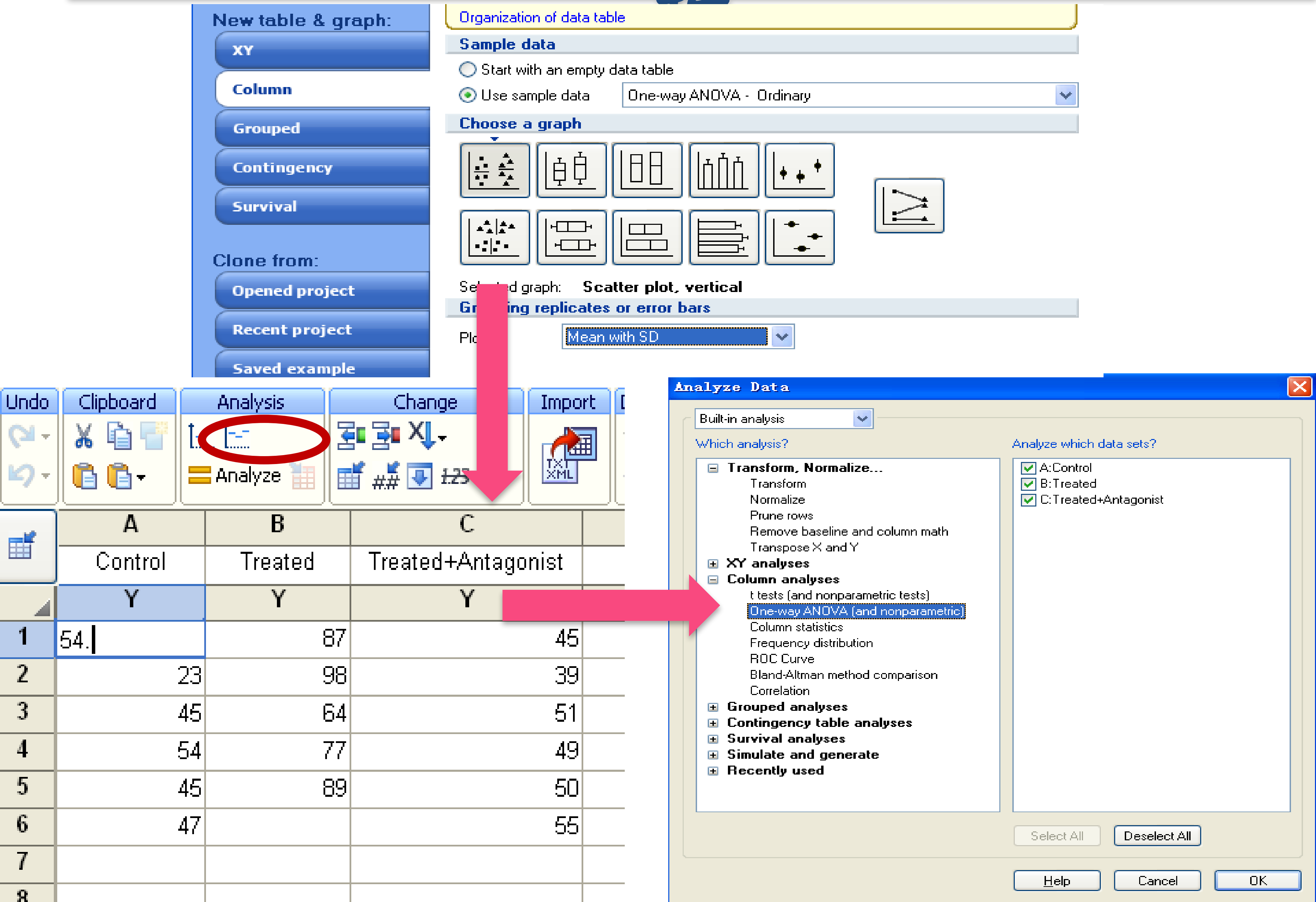
P>0.05

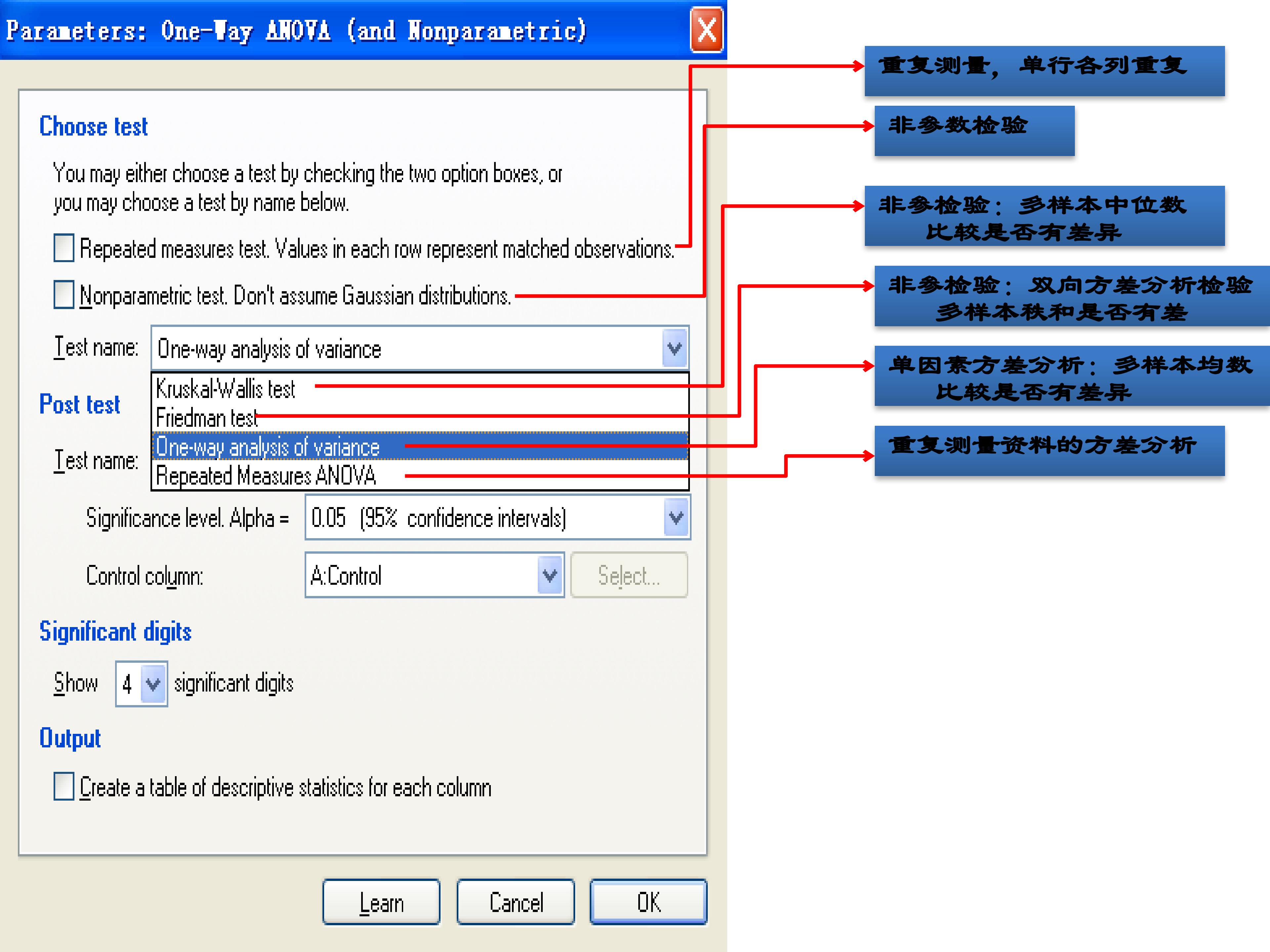
差异无统计学意义



*
YTitle
0
20
40
60
80
100
Before
After

One-way anova （单因素方差）分析




Parameters: One-Way ANOVA (and Nonparametric)
Choose test
You may either choose a test by checking the two option boxes, or you may choose a test by name below.
Repeated measures test. Values in each row represent matched observations.
Nonparametric test. Don't assume Gaussian distributions.
Test name:
One-way analysis of variance
Kruskal-Wallis test
Friedman test
One-way analysis of variance
Repeated Measures ANOVA
Post test
Test name:
Significance level. Alpha =
0.05 (95% confidence intervals)
Control column:
A:Control
Select...
Significant digits
Show
4
significant digits
Output
Create a table of descriptive statistics for each column
Learn
Cancel
OK
重复测量，单行各列重复
非参数检验
非参检验：多样本中位数比较是否有差异
非参检验：双向方差分析检验多样本秩和是否有差
单因素方差分析：多样本均数比较是否有差异
重复测量资料的方差分析

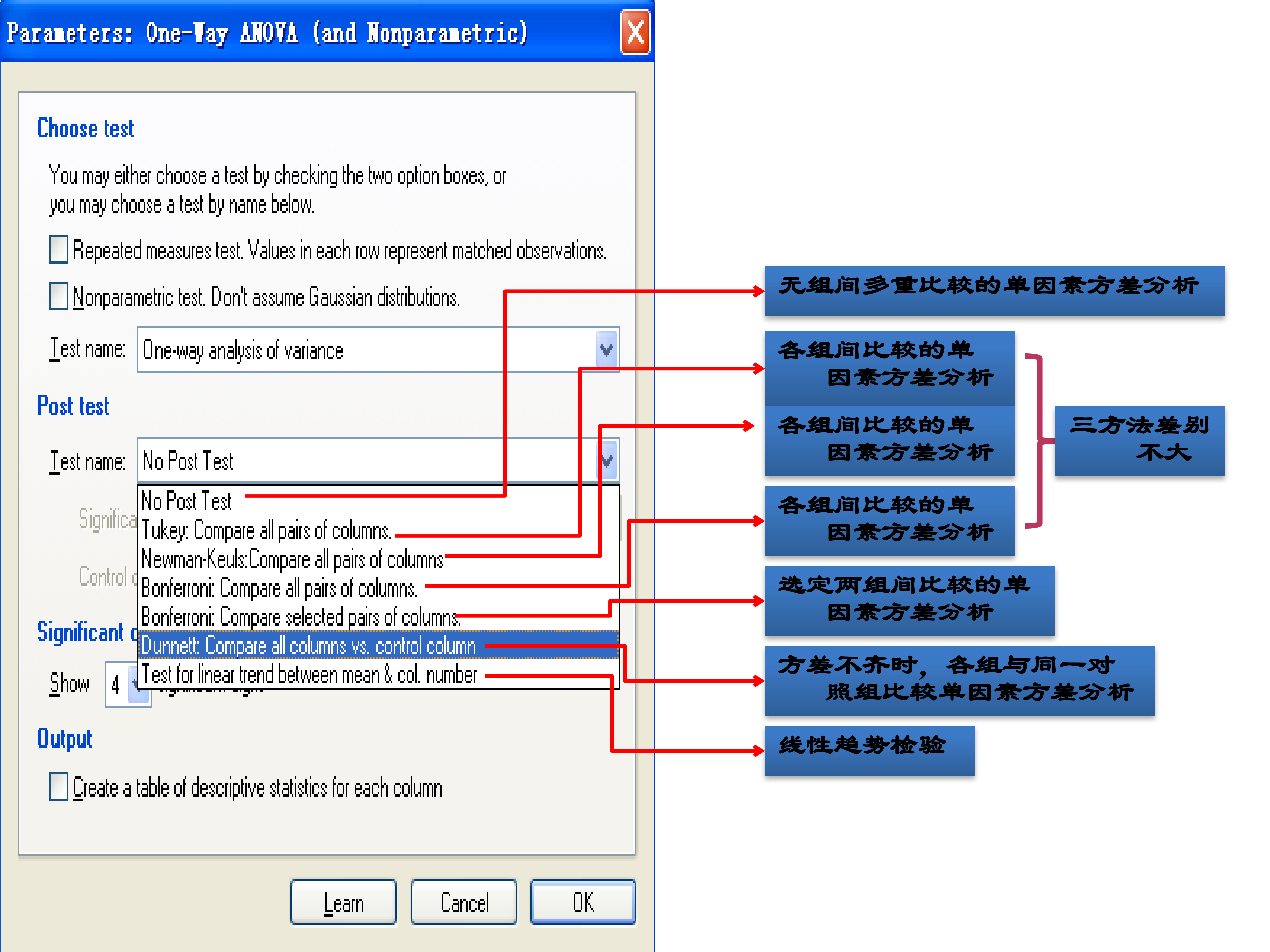

Parameters: One-Way ANOVA (and Nonparametric)
Choose test
You may either choose a test by checking the two option boxes, or
you may choose a test by name below.
Repeated measures test. Values in each row represent matched observations.
Nonparametric test. Don't assume Gaussian distributions.
Test name: One-way analysis of variance
Post test
Test name: No Post Test
No Post Test
Tukey: Compare all pairs of columns.
Newman-Keuls:Compare all pairs of columns
Bonferroni: Compare all pairs of columns.
Bonferroni: Compare selected pairs of columns.
Dunnett: Compare all columns vs. control column
Test for linear trend between mean & col. number
Significa
Control c
Significant d
Show 4
Output
Create a table of descriptive statistics for each column
Learn
Cancel
OK
无组间多重比较的单因素方差分析
各组间比较的单因素方差分析
各组间比较的单因素方差分析
各组间比较的单因素方差分析
三方法差别不大
选定两组间比较的单因素方差分析
方差不齐时，各组与同一对照组比较单因素方差分析
线性趋势检验

Table Analyzed	One-way ANOVA data		
One-way analysis of variance			
P value	P<0.0001		
P value summary	***		
Are means signif. different? (P < 0.05)	Yes		
Number of groups	3		
F	22.57		
R squared	0.7633		
Bartlett's test for equal variances			
Bartlett's statistic (corrected)	2.986		
P value	0.2247		
P value summary	ns		
Do the variances differ signif. (P < 0.05)	No		
ANOVA Table	SS	df	MS
Treatment (between columns)	4760	2	2380
Residual (within columns)	1476	14	105.4
Total	6236	16	
Tukey's Multiple Comparison Test	Mean Diff.	q	Significant? P < 0.05?
Control vs Treated	-38.33	8.719	Yes
Control vs Treated+Antagonist	-3.500	0.8349	No
Treated vs Treated+Antagonist	34.83	7.923	Yes

P<0.01

差异有统计学意义

统计量 F=22.57

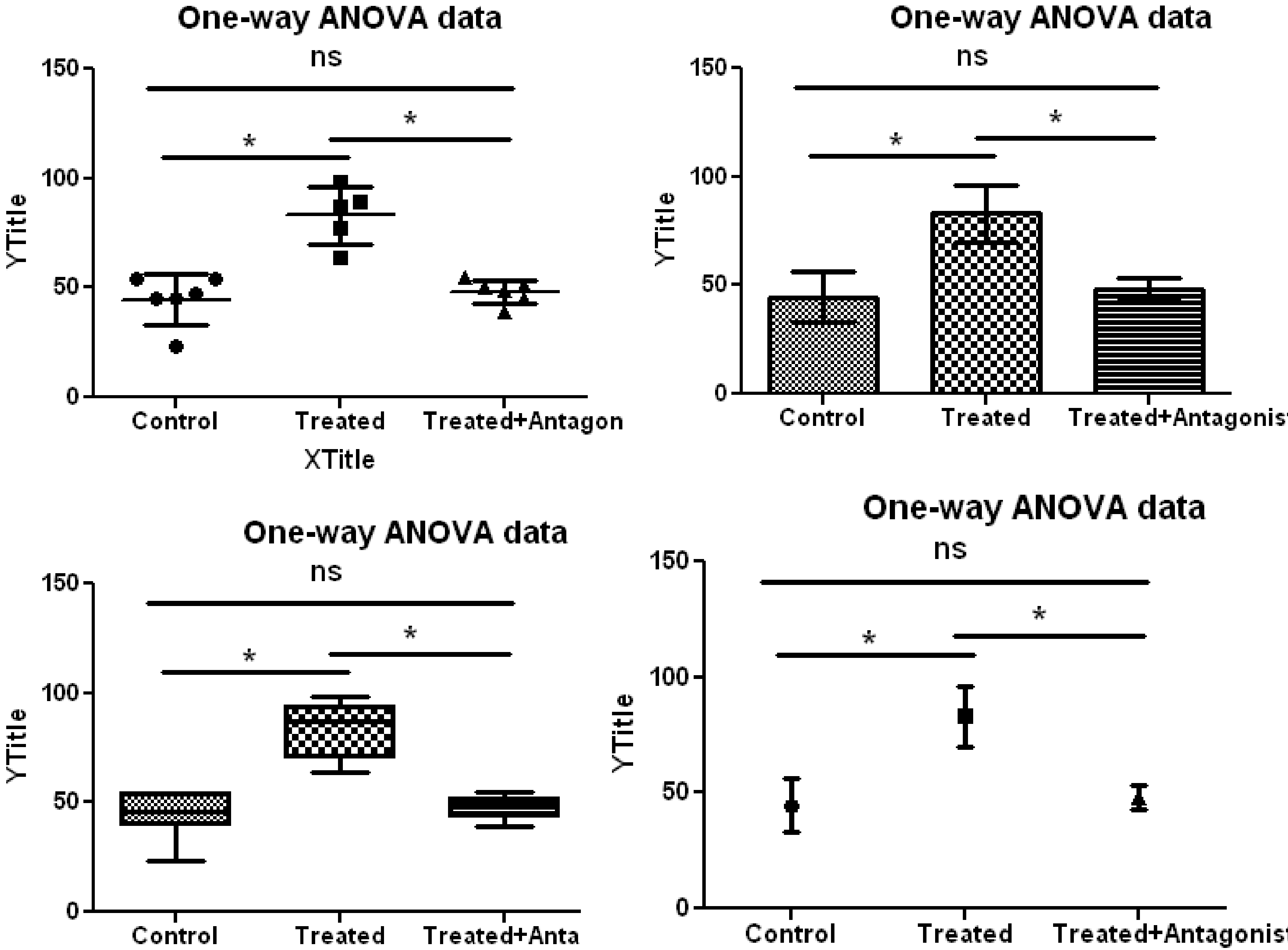
Bartlett’s 方差齐性检验

方差齐性检验 p>0.05

Bartlett’s 检验方差齐

Tukey’s 多重比较差异有统计学意义

Control 与 Treated+Ant 组比较差异无统计学意义


One-way ANOVA data
ns
*
*
150
100
50
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YTitle
Control
Treated
Treated+Antagon
XTitle
One-way ANOVA data
ns
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YTitle
Control
Treated
Treated+Antagonist
One-way ANOVA data
ns
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150
100
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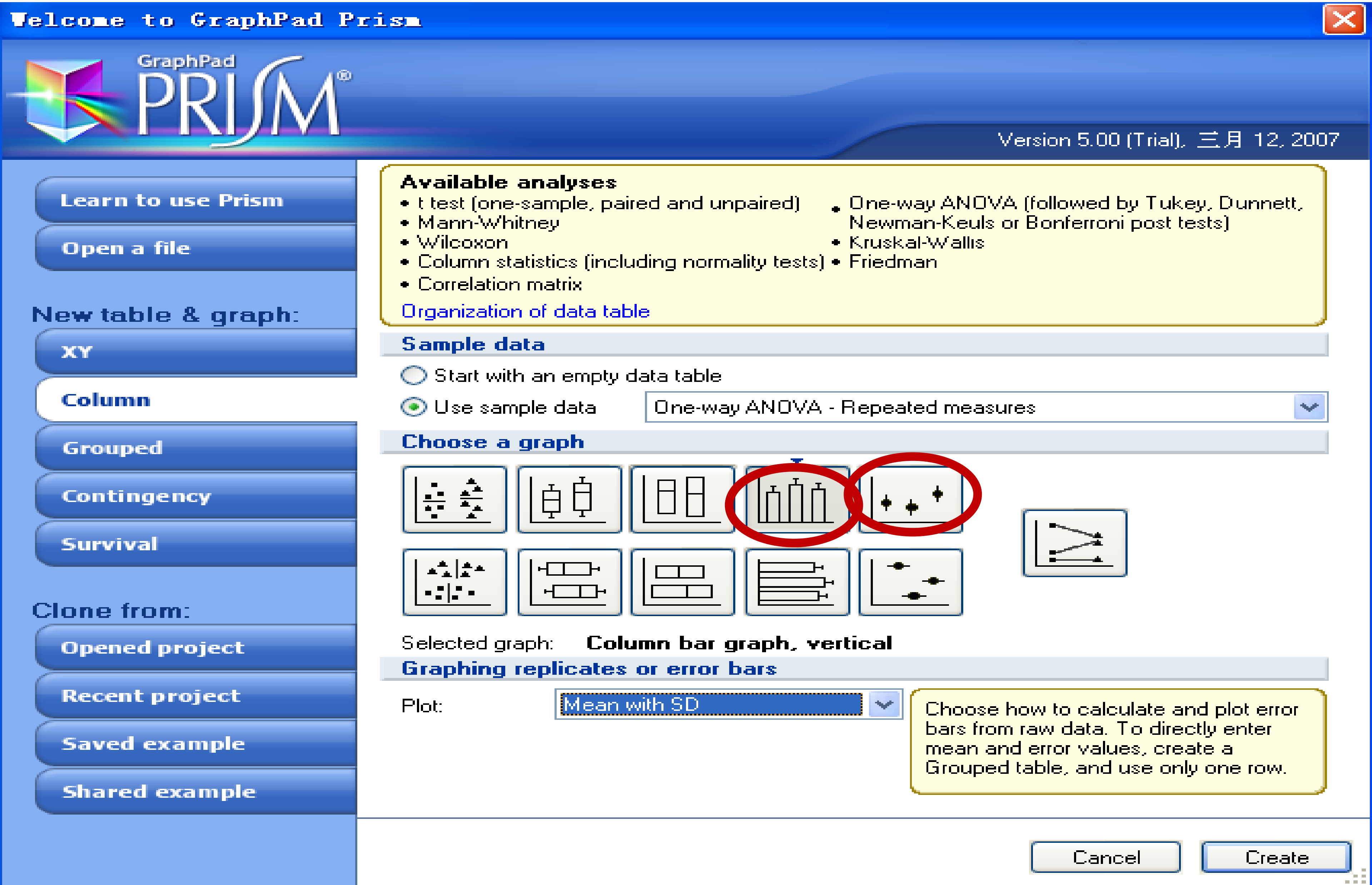
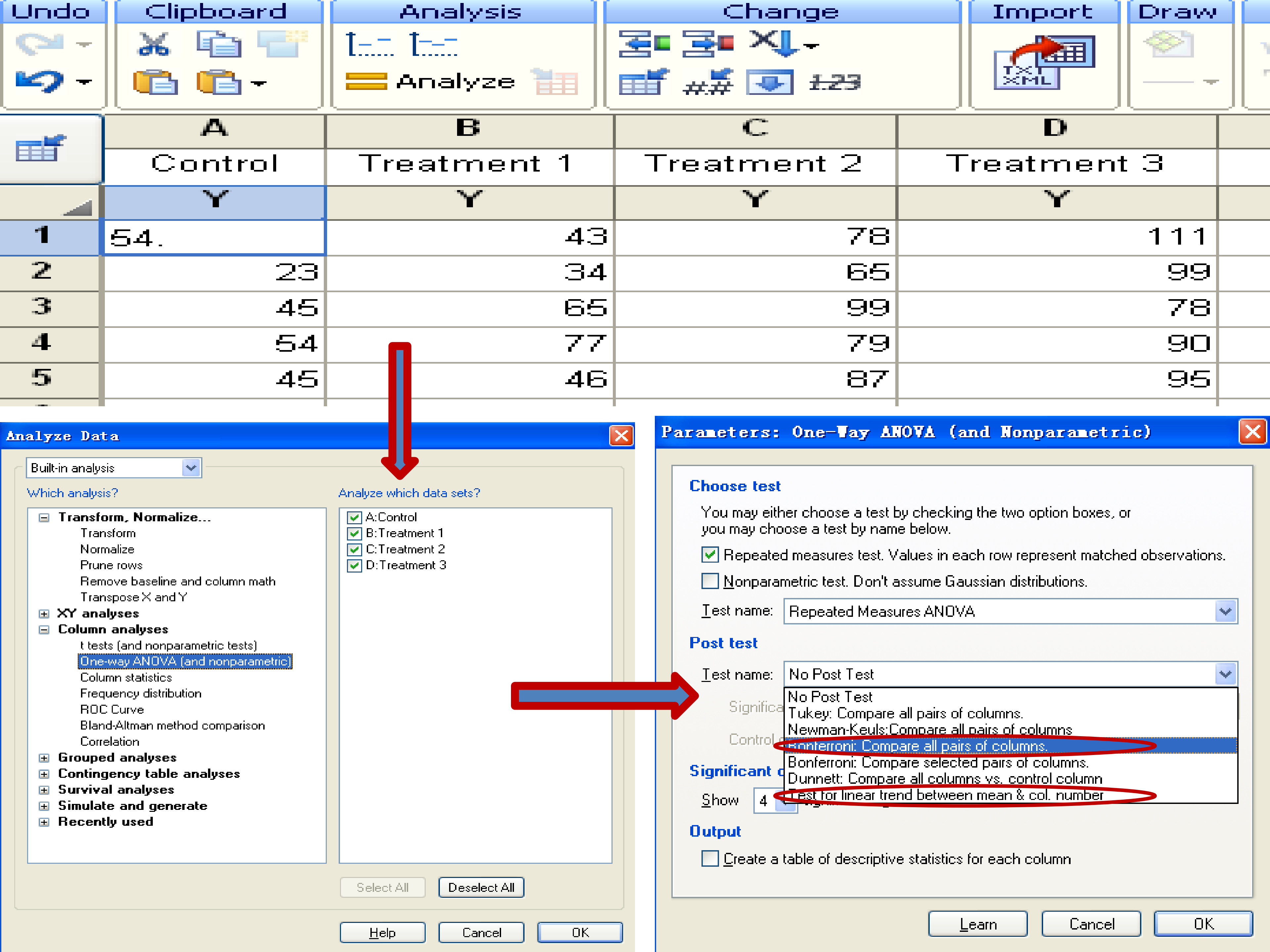
YTitle
Control
Treated
Treated+Anta
One-way ANOVA data
ns
*
*
150
100
50
0
YTitle
Control
Treated
Treated+Antagonist

One-way anova－Repeated measures （单因素方差分析－重复测量）

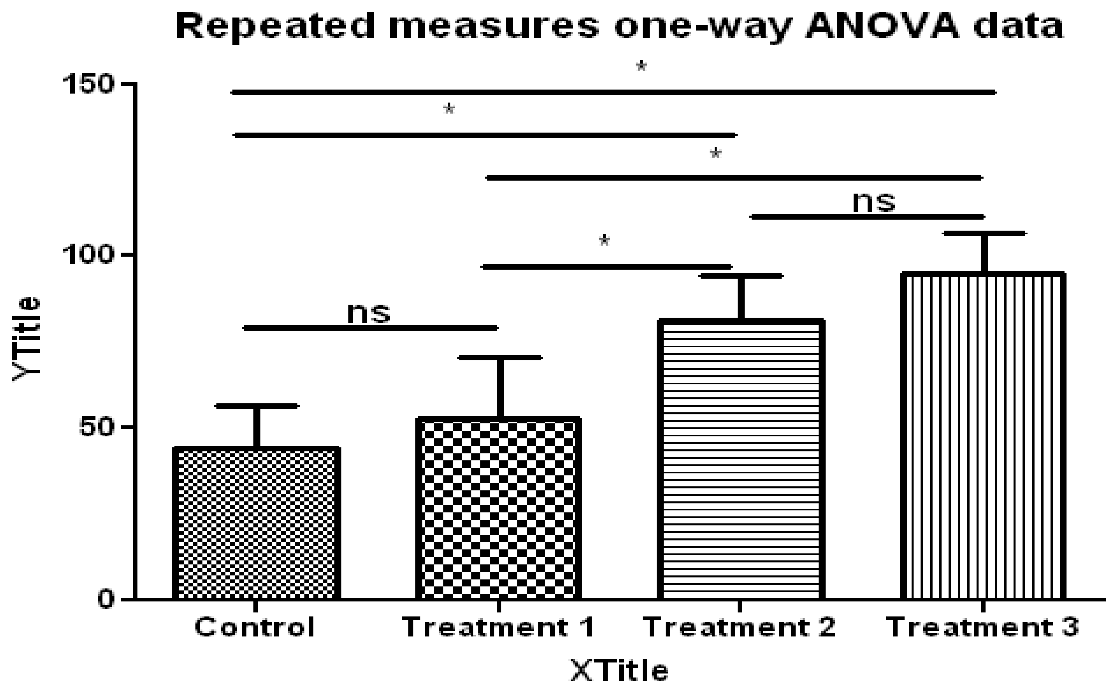




Undo
Clipboard
Analysis
Change
Import
Draw
Analyze
A
B
C
D
Control
Treatment 1
Treatment 2
Treatment 3
Y
1 54. 43 78 111
2 23 34 65 99
3 45 65 99 78
4 54 77 79 90
5 45 46 87 95
Analyze Data
Built-in analysis
Which analysis?
Analyze which data sets?
Transform, Normalize...
Transform
Normalize
Prune rows
Remove baseline and column math
Transpose X and Y
XY analyses
Column analyses
t tests (and nonparametric tests)
One-way ANOVA (and nonparametric)
Column statistics
Frequency distribution
ROC Curve
Bland-Altman method comparison
Correlation
Grouped analyses
Contingency table analyses
Survival analyses
Simulate and generate
Recently used
A:Control
B:Treatment 1
C:Treatment 2
D:Treatment 3
Select All
Deselect All
Help
Cancel
OK
Parameters: One-Way ANOVA (and Nonparametric)
Choose test
You may either choose a test by checking the two option boxes, or
you may choose a test by name below.
Repeated measures test. Values in each row represent matched observations.
Nonparametric test. Don't assume Gaussian distributions.
Test name:
Repeated Measures ANOVA
Post test
Test name:
No Post Test
No Post Test
Tukey: Compare all pairs of columns.
Newman-Keuls:Compare all pairs of columns
Bonferroni: Compare all pairs of columns.
Bonferroni: Compare selected pairs of columns.
Dunnett: Compare all columns vs. control column
Test for linear trend between mean & col. number
Significa
Control
Significant
Show
4
Output
Create a table of descriptive statistics for each column
Learn
Cancel
OK

Repeated Measures ANOVA			
P value	0.0002		
P value summary	***		
Are means signif. different? (P < 0.05)	Yes		
Number of groups	4		
F	15.76		
R squared	0.7976		
Was the pairing significantly effective?			
R squared	0.08253		
F	1.333		
P value	0.3134		
P value summary	ns		
Is there significant matching? (P < 0.05)	No		
ANOVA Table	SS	df	MS
Treatment (between columns)	8417	3	2806
Individual (between rows)	949.3	4	237.3
Residual (random)	2136	12	178.0
Total	11500	19	
Bonferroni's Multiple Comparison Test	Mean Diff.	t	Significant? P < 0.05?
Control vs Treatment 1	-8.800	1.043	No
Control vs Treatment 2	-37.40	4.432	Yes
Control vs Treatment 3	-50.40	5.973	Yes
Treatment 1 vs Treatment 2	-28.60	3.390	Yes
Treatment 1 vs Treatment 3	-41.60	4.930	Yes
Treatment 2 vs Treatment 3	-13.00	1.541	No

1	Table Analyzed	Repeated measures one-way ANOVA data		
2				
3	Repeated Measures ANOVA			
4	P value	0.0002		
5	P value summary	***		
6	Are means signif. different? (P < 0.05)	Yes		
7	Number of groups	4		
8	F	15.76		
9	R squared	0.7976		
10				
11	Was the pairing significantly effective?			
12	R squared	0.08253		
13	F	1.333		
14	P value	0.3134		
15	P value summary	ns		
16	Is there significant matching? (P < 0.05)	No		
17				
18	ANOVA Table	SS	df	MS
19	Treatment (between columns)	8417	3	2806
20	Individual (between rows)	949.3	4	237.3
21	Residual (random)	2136	12	178.0
22	Total	11500	19	
23				
24	Post test for linear trend			
25	Slope	8.990		
26	R squared	0.7026		
27	P value	P<0.0001		
28	P value summary	***		
29	Is linear trend significant (P < 0.05)?	Yes		


Repeated measures one-way ANOVA data
150
100
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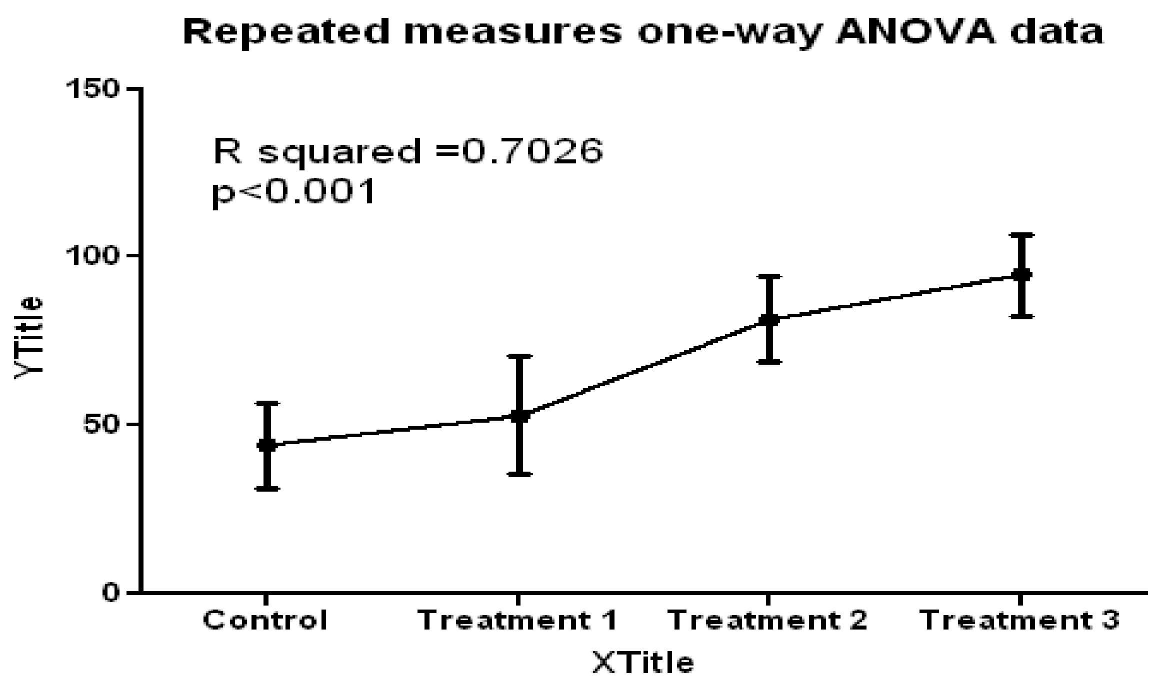
YTitle
ns
*
*
*
ns
*
Control
Treatment 1
Treatment 2
Treatment 3
XTitle


Repeated measures one-way ANOVA data
R squared =0.7026
p<0.001
150
100
50
0
YTitle
Control
Treatment 1
Treatment 2
Treatment 3
XTitle